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MODULE IV

LAPLACE TRANSFORMS

Topics :

- * Laplace transforms and its inverse
- * Existence theorem without proof
- * Linearity
- * Laplace transforms of basic function
- * 1st shifting theorem
- * LT of derivatives and integrals
- * Solution of differential equation using LT
- * Unit step function
- * 2nd shifting Theorem
- * Dirac Delta Function and its LT
- * Solution of ordinary differential equation involving unit
- * Convolution without proof and its application to find inverse laplace transforms of products of function

Laplace Transforms if $f(t)$ is a function defined for all $t \geq 0$, its laplace transform is defined as ;

$$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$L\{f(t)\} = F(s)$$

Note 1 : original function depends of 't' and their transforms on 's'

Note 2 : LT is called as integral transform because it transforms a function into another by a process of integration.

LINEARITY OF LT

For any two functions $f(t)$ and $g(t)$ whose laplace transform exist and 'a' and 'b' are any constants;

$$L\{af(t) + bg(t)\} = aL\{f(t)\} + bL\{g(t)\}$$

proof :

$$L\{af(t) + bg(t)\} = \int_0^{\infty} e^{-st} (af(t) + bg(t)) dt$$

$$= \int_0^{\infty} a e^{-st} f(t) dt + \int_0^{\infty} b e^{-st} g(t) dt$$

$$= a \int_0^{\infty} e^{-st} f(t) dt + b \int_0^{\infty} e^{-st} g(t) dt$$

$$= a L\{f(t)\} + b L\{g(t)\}$$

LAPLACE TRANSFORMS OF BASIC FUNCTIONS

1. $L\{1\} = \frac{1}{s}$

proof :

$$L\{1\} = \int_0^{\infty} e^{-st} \cdot 1 \cdot dt$$

$$= \int_0^{\infty} e^{-st} dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_0^{\infty}$$

$$= \frac{e^{-\infty} - e^0}{-s} = \frac{1}{s}$$

2. $L\{e^{at}\} = \frac{1}{s-a}$

proof :

$$L\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt$$

$$= \int_0^{\infty} e^{-st+at} dt$$

$$= \int_0^{\infty} e^{-t(s-a)} dt$$

$$= \left[\frac{e^{-t(s-a)}}{-(s-a)} \right]_0^{\infty} = \frac{e^{-\infty} - e^0}{-(s-a)} = \frac{1}{s-a}$$

$$3. \quad \boxed{L\{e^{-at}\} = \frac{1}{s+a}}$$

proof :

$$L\{e^{-at}\} = \int_0^{\infty} e^{-st} \cdot e^{-at} dt$$

$$= \int_0^{\infty} e^{-st-at} dt$$

$$= \int_0^{\infty} e^{-t(a+s)} dt$$

$$= \left[\frac{e^{-t(a+s)}}{-(a+s)} \right]_0^{\infty}$$

$$= \frac{e^{-\infty} - e^0}{-(a+s)}$$

$$= \frac{0 - 1}{-(a+s)} = \frac{1}{a+s}$$

$$4. \quad \boxed{L\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}}$$

proof :

$$L\{\sin \omega t\} = \int_0^{\infty} e^{-st} \sin \omega t dt$$

$$= \left[\frac{e^{-st}}{s^2 + \omega^2} [-s \sin \omega t - \omega \cos \omega t] \right]_0^{\infty}$$

$$= \frac{1}{s^2 + \omega^2} \left[e^{\dots} - e^0 [-s \sin 0 - \omega \cos 0] \right]$$

$$= \frac{\omega}{s^2 + \omega^2}$$

5. $\boxed{L\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}}$

proof :

$$L\{\cos \omega t\} = \int_0^{\infty} e^{-st} \cos \omega t \, dt$$

$$= \left[\frac{e^{-st}}{s^2 + \omega^2} [-s \cos \omega t + \omega \sin \omega t] \right]_0^{\infty}$$

$$= \frac{1}{s^2 + \omega^2} \left[e^{-\infty} (-s \cos \infty + \omega \sin \infty) - e^0 [-s \cos 0 + \omega \sin 0] \right]$$

$$= \frac{s}{s^2 + \omega^2}$$

6. $L\{\sinh \omega t\} = \frac{\omega}{s^2 - \omega^2}$

proof :

$$L\{\sinh \omega t\} = L\left\{ \frac{e^{\omega t} - e^{-\omega t}}{2} \right\}$$

$$= \frac{1}{2} [L\{e^{\omega t}\} - L\{e^{-\omega t}\}]$$

$$= \frac{1}{2} \left[\frac{1}{s - \omega} - \frac{1}{s + \omega} \right]$$

$$= \frac{1}{2} \left[\frac{s+\omega - s+\omega}{s^2 + \omega^2} \right]$$

$$= \frac{\omega}{s^2 + \omega^2}$$

$$7. \quad L\{\cosh \omega t\} = \frac{s}{s^2 + \omega^2}$$

Proof:

$$L\{\cosh \omega t\} = L\left\{ \frac{e^{\omega t} + e^{-\omega t}}{2} \right\}$$

$$= \frac{1}{2} [L\{e^{\omega t}\} + L\{e^{-\omega t}\}]$$

$$= \frac{1}{2} \left[\frac{1}{s-\omega} + \frac{1}{s+\omega} \right]$$

$$= \frac{1}{2} \left[\frac{s+\omega + s-\omega}{s^2 + \omega^2} \right]$$

$$= \frac{s}{s^2 + \omega^2}$$

$$8. \quad L\{t\} = \frac{1!}{s^2}$$

proof:

$$L\{t\} = \int_0^{\infty} e^{-st} \cdot t \cdot dt \quad [\text{Integral} \cdot \text{UV Rule}]$$

$$= \int_0^{\infty} t \cdot e^{-st} dt$$

$$= \left[(t) \left(\frac{e^{-st}}{-s} \right) - (1) \left(\frac{e^{-st}}{s^2} \right) + 0 \right]_0^{\infty}$$

$$= \left(t \cdot \frac{e^{-st}}{-s} - \frac{e^{-st}}{s^2} \right) - \left(0 \cdot \frac{e^0}{-s} - \frac{e^0}{s^2} \right)$$

$$= \frac{1}{s^2}$$

9. $L\{t^2\} = \frac{2!}{s^3}$

proof:

$$L\{t^2\} = \int_0^{\infty} e^{-st} \cdot t^2 \cdot dt$$

$$= \int_0^{\infty} t^2 \cdot e^{-st} dt$$

$$= \left[(t^2) \cdot \left(\frac{e^{-st}}{-s} \right) - (2t) \cdot \left(\frac{e^{-st}}{s^2} \right) + (2) \cdot \left(\frac{e^{-st}}{-s^3} \right) \right]_0^{\infty}$$

$$= \left(0 \cdot \frac{e^{-\infty}}{-s} - 2 \cdot \frac{e^{-\infty}}{s^2} + 2 \cdot \frac{e^{-\infty}}{-s^3} \right) - \left(0 - 0 + 2 \cdot \frac{e^0}{-s^3} \right)$$

$$= \frac{2!}{s^3}$$

ie; $L\{t\} = \frac{1}{s^2}$

$$L\{t^2\} = \frac{2!}{s^3}$$

$$L\{t^3\} = \frac{3!}{s^4}$$

⋮

In general $L\{t^n\} = \frac{n!}{s^{n+1}}$ by mathematical Induction.

$$10. \quad L\{t^n\} = \frac{n!}{s^{n+1}}$$

FIRST SHIFTING THEOREM (s-shifting)

OR (Multiplication by e^{at})

$$L\{e^{at} f(t)\} = \int_0^{\infty} e^{-st} \cdot e^{at} f(t) dt$$

$$= \int_0^{\infty} e^{-st+at} f(t) dt$$

$$= \int_0^{\infty} e^{-(s-a)t} f(t) dt$$

$$= F(s-a)$$

since.

$$\int_0^{\infty} e^{-st} f(t) dt = F(s)$$

~~$$\int_0^{\infty} e^{-(s-a)t} f(t) dt = F(s-a)$$~~

$$L\{e^{at} f(t)\} = F(s-a), \quad s \rightarrow s-a$$

$$L\{e^{-at} f(t)\} = F(s+a), \quad s \rightarrow s+a$$

Q. Find $L\{2t+8\}$

$$\begin{aligned}\text{Ans: } L\{2t+8\} &= L\{2t\} + L\{8\} \\ &= 2L\{t\} + 8L\{1\} \\ &= 2 \times \frac{1}{s^2} + \frac{8}{s} \\ &= \frac{2}{s^2} + \frac{8}{s}\end{aligned}$$

(ii) $L\{\cos 2\pi t\}$

$$\text{Ans: } L\{\cos 2\pi t\} = \frac{s}{s^2 + 4\pi^2}$$

$$\begin{aligned}\text{(iii) } L\{(a-b)^2\} &= L\{a^2 + b^2t^2 - 2abt\} \\ &= L\{a^2\} + L\{b^2t^2\} - L\{2abt\} \\ &= a^2L\{1\} + b^2L\{t^2\} - 2abL\{t\} \\ &= \frac{a^2}{s} + \frac{2b^2}{s^3} - \frac{2ab}{s^2}\end{aligned}$$

$$\begin{aligned}\text{(iv) } L\{\cos^2 \omega t\} &= L\left\{\frac{1 + \cos 2\omega t}{2}\right\} \\ &= \frac{1}{2} [L\{1\} + L\{\cos 2\omega t\}] \\ &= \frac{1}{2s} + \frac{s}{2(s^2 + 4\omega^2)}\end{aligned}$$

$$(v) \mathcal{L}\{\cos(\omega t + \theta)\}$$

$$\begin{aligned} \text{Ans: } \mathcal{L}\{\cos(\omega t + \theta)\} &= \mathcal{L}\{\cos \omega t \cos \theta - \sin \omega t \sin \theta\} \\ &= \cos \theta \times \left(\frac{s}{s^2 + \omega^2} \right) - \sin \theta \left(\frac{\omega}{s^2 + \omega^2} \right) \end{aligned}$$

$$(vi) \mathcal{L}\{\sin^2 t\}$$

$$= \mathcal{L}\left\{ \frac{1 - \cos 2t}{2} \right\}$$

$$= \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 4} \right]$$

$$(vii) \mathcal{L}\{\sin t \cos 2t\}$$

$$= \mathcal{L}\left\{ \frac{\sin(t+2t) + \sin(t-2t)}{2} \right\}$$

$$= \mathcal{L}\left\{ \frac{\sin(3t) - \sin(t)}{2} \right\}$$

$$= \frac{1}{2} \left[\frac{3}{s^2 + 9} - \frac{1}{s^2 + 1} \right]$$

$$(viii) \mathcal{L}\{\cos \sqrt{t}\}$$

$$(ix) \mathcal{L}\{e^{at} \cos \omega t\}$$

$$= \frac{s-a}{(s-a)^2 + \omega^2}$$

$$(x) \mathcal{L}\{e^{3t} \sinh t\}$$

$$= \frac{1}{(s-3)^2 - 1}$$

$$(xi) \mathcal{L}\{e^{-t} \cdot \sinh 4t\}$$

$$= \frac{4}{(s+1)^2 - 4^2}$$

$$(xii) \mathcal{L}\{(1.5) \sin(3t - \frac{\pi}{2})\}$$

$$= (1.5) \mathcal{L}\{\sin(3t - \frac{\pi}{2})\}$$

$$= (1.5) \mathcal{L}\{\sin 3t \cos \frac{\pi}{2} - \cos 3t \sin \frac{\pi}{2}\}$$

$$= (1.5) \mathcal{L}\{-\cos 3t\}$$

$$= -(1.5) \times \frac{s}{s^2 + 9}$$

12) $L\{\cosh^2 2t\}$

13) $L\{t^3 \cdot e^{-2t}\}$

by 1st shifting theorem

$$= \frac{3!}{(s+2)^4}$$

14) $L\{k e^{-at} \cos \omega t\}$

$$= k L\{e^{-at} \cos \omega t\}$$

By 1st shifting theorem

$$= k \times \frac{(s+a)}{(s+a)^2 + \omega^2}$$

15) $L\{2 e^{-1/2 t} \sin 4\pi t\}$

$$= 2 L\{e^{-1/2 t} \sin 4\pi t\}$$

By 1st shifting theorem

$$= 2 \times \frac{4\pi}{(s+\frac{1}{2})^2 + 16\pi^2}$$

(imp)

7) $L\{\cos(\sqrt{t})\}$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\therefore \cos \sqrt{t} = 1 - \frac{t}{2!} + \frac{t^2}{4!} - \frac{t^3}{6!} + \dots$$

Ans

$$\mathcal{L}\{\cos Jt\} = \frac{1}{s} - \frac{1}{2!s^2} + \frac{2}{4!s^3} - \frac{3}{6!s^4} - \dots$$

$$\begin{aligned} 16) \mathcal{L}\{\sin ht \cos t\} &= \mathcal{L}\left\{\frac{\sin(ht+t) + \sin(ht-t)}{2}\right\} \\ &= \frac{1}{2} \mathcal{L}\{\sin(h+1)t + \sin(h-1)t\} \end{aligned}$$

imp
16) $\mathcal{L}\{\cosh^2 2t\}$

$$\cosh 2t = \frac{e^{2t} + e^{-2t}}{2}$$

$$\cosh^2 2t = \frac{e^{4t} + e^{-4t} + 2e^{2t-2t}}{4}$$

$$= \frac{e^{4t} + e^{-4t} + 2}{4}$$

$$\therefore \mathcal{L}\left\{\frac{e^{4t} + e^{-4t} + 2}{4}\right\}$$

$$= \frac{1}{4} \left[\frac{1}{s-4} + \frac{1}{s+4} + \frac{2}{s} \right]$$

$$17) \mathcal{L}\{\sinh^2 2t\}$$

$$\sinh 2t = \frac{e^{2t} - e^{-2t}}{2}$$

$$\sinh^2 2t = \frac{e^{4t} + e^{-4t} - 2}{4}$$

$$\therefore L \left\{ \frac{e^{4t} + e^{-4t} - 2}{4} \right\}$$

$$= \frac{1}{4} \left[\frac{1}{s-4} + \frac{1}{s+4} - \frac{2}{s} \right]$$

18) $L \{ \sinh t \cdot \cos t \}$

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

$$\therefore L \left\{ \frac{(e^t + e^{-t}) \cos t}{2} \right\}$$

$$= \frac{1}{2} [L \{ e^t \cos t \} + L \{ e^{-t} \cos t \}]$$

$$= \frac{1}{2} \left[\frac{s-1}{(s-1)^2 + 1} + \frac{s+1}{(s+1)^2 + 1} \right]$$

19) $L \{ \sin^4 t \}$

$$\sin^2 t = \frac{1 - \cos 2t}{2}$$

$$\therefore \sin^4 t = \frac{(1 - \cos 2t)^2}{4} = \frac{1 + \cos^2 2t - 2 \cos 2t}{4}$$

$$= \frac{1}{4} \left[1 + \frac{1 + \cos 4t}{2} - 2 \cos 2t \right]$$

$$= \frac{1}{4} \left[1 + \frac{1}{2} + \frac{\cos 4t}{2} - 2 \cos 2t \right]$$

$$\therefore L\{\sin^4 t\} = \frac{1}{4} \left[\frac{1}{s} + \frac{1}{2s} + \frac{s}{2(s^2+16)} - \frac{2s}{s^2+4} \right]$$

imp

20. Find the Laplace transform of $f(t)$ where $f(t)$

$$f(t) = \begin{cases} t & ; 0 < t < 1 \\ 1 & ; t > 1 \end{cases}$$

Ans: $L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$

$$= \int_0^1 e^{-st} t dt + \int_1^{\infty} e^{-st} \cdot 1 dt$$

$$= \left[(t) \left(\frac{e^{-st}}{-s} \right) - (1) \left(\frac{e^{-st}}{s^2} \right) \right]_0^1 + \left[\frac{e^{-st}}{-s} \right]_1^{\infty}$$

$$= \frac{-s}{-s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} + \frac{e^{-s}}{s}$$

$$= \frac{1 - e^{-s}}{s^2}$$

21) Find Laplace transform $f(t) = \begin{cases} 1 & 0 < t < 1 \\ -1 & 1 < t < 2 \end{cases}$

Ans: $L\{f(t)\} = \int_0^1 e^{-st} \cdot 1 dt + \int_1^2 e^{-st} \cdot (-1) dt$

$$= \int_0^1 e^{-st} dt - \int_1^2 e^{-st} dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_0^1 - \left[\frac{e^{-st}}{-s} \right]_1^2$$

$$= -\left(\frac{e^{-s}}{s} + \frac{e^{-s}}{s} \right) + \frac{e^{-2s}}{s} - \frac{e^{-s}}{s}$$

$$= \frac{-e^{-s}}{-s} + \frac{1}{s} + \frac{e^{-2s}}{s} - \frac{e^{-s}}{s}$$

$$= \frac{1 + e^{-2s} - 2e^{-s}}{s}$$

INVERSE LAPLACE TRANSFORM

If $L\{f(t)\} = F(s)$

then the given function $f(t)$ is called the inverse transform of $F(s)$ and is denoted by ~~$f(t)$~~

$$L^{-1}\{F(s)\} = f(t)$$

Laplace Transform	Inverse Laplace transform
$L\{1\} = \frac{1}{s}$	$L^{-1}\left\{\frac{1}{s}\right\} = 1$
$L\{e^{at}\} = \frac{1}{s-a}$	$L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$
$L\{e^{-at}\} = \frac{1}{s+a}$	$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$
$L\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$	$L^{-1}\left\{\frac{s}{s^2 + \omega^2}\right\} = \cos \omega t$
$L\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$	$L^{-1}\left\{\frac{\omega}{s^2 + \omega^2}\right\} = \sin \omega t$
$L\{\cosh \omega t\} = \frac{s}{s^2 - \omega^2}$	$L^{-1}\left\{\frac{1}{s^2 + \omega^2}\right\} = \frac{\sin \omega t}{\omega}$

$$\mathcal{L}\{\sinh \omega t\} = \frac{\omega}{s^2 - \omega^2}$$

~~$$\mathcal{L}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$$~~

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}^{-1}\left\{\frac{\omega}{s^2 - \omega^2}\right\} = \sinh \omega t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 - \omega^2}\right\} = \frac{\sinh \omega t}{\omega}$$

$$\mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\} = t^n$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$$

~~Reverse~~
First shifting theorem

Inverse of 1st shifting theorem

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$$

$$\mathcal{L}\{e^{-at} f(t)\} = F(s+a)$$

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at} f(t)$$

$$\mathcal{L}^{-1}\{F(s+a)\} = e^{-at} f(t)$$

Q. 1. find $f(t)$ where $F(s) = \frac{0.2s + 1.4}{s^2 + 1.96}$

Ans: If in Question 's' is present then we ~~do~~ $\mathcal{L}^{-1}\{F(s)\}$

$$\therefore F(s) = \frac{0.2s}{s^2 + 1.96} + \frac{1.4}{s^2 + 1.96}$$

$$= (0.2) \cdot \mathcal{L}^{-1}\left\{\frac{s}{s^2 + (1.4)^2}\right\} + \mathcal{L}^{-1}\left\{\frac{(1.4)}{s^2 + (1.4)^2}\right\}$$

$$= \underline{\underline{0.2 \cos(1.4t) + \sin(1.4t)}}$$

2) $F(s) = \frac{5s+1}{s^2-25}$

Ans: $F(s) = \frac{5s}{s^2-25} + \frac{1}{s^2-25}$

$= \cosh(s) \left(\frac{s}{s^2-(s)^2} \right) + \frac{1}{s^2-s^2}$

$\mathcal{L}^{-1}(F(s)) = \underline{\underline{5 \cosh st + \sinh st}}$

3) $F(s) = \frac{s}{L^2 s^2 + \left(\frac{\pi}{2L}\right)^2}$

Ans: $F(s) = \frac{s}{L^2 \left(s^2 + \left(\frac{\pi}{2L}\right)^2 \right)}$

$\mathcal{L}^{-1} F(s) = \underline{\underline{\frac{1}{L^2} \cos \frac{\pi}{2L} t}}$

11.10

Q. $f(t) = 1 - \frac{t}{2} \quad 0 < t < 1$

Ans: $\mathcal{L}\{f(t)\} = \int_0^1 e^{-st} \left(1 - \frac{t}{2}\right) dt$
 $= \int_0^1 e^{-st} dt - \frac{1}{2} \int_0^1 e^{-st} \cdot t dt$

$$= \left[\frac{e^{-st}}{-s} \right]_0^1 - \frac{1}{2} \left[(t) \left(\frac{e^{-st}}{-s} \right) - (1) \left(\frac{e^{-st}}{s^2} \right) \right]_0^1$$

$$= \frac{e^{-s}}{-s} + \frac{1}{-s} - \frac{1}{2} \left[\frac{e^{-s}}{-s} + \frac{e^{-s}}{-s^2} - \frac{1}{s^2} \right]$$

$$= \frac{e^{-s}}{-s} + \frac{1}{-s} - \frac{1}{2} \left[\frac{2e^{-s}}{-s} - \frac{1}{s^2} \right]$$

$$= \frac{\cancel{e^{-s}}}{-s} + \frac{1}{-s} + \frac{\cancel{e^{-s}}}{\cancel{-s}} + \frac{1}{2s^2}$$

$$= \underline{\underline{\frac{1}{2s^2} - \frac{1}{s}}}$$

Q. Find L^{-1} of $F(s) = \frac{1}{(s+\sqrt{2})(s+\sqrt{3})}$

Ans:

$$F(s) = \frac{1}{s^2 + s\sqrt{3} + s\sqrt{2} + \sqrt{6}}$$

$$= \frac{1}{s^2 + s(\sqrt{2} + \sqrt{3}) + \sqrt{6}}$$

$$\frac{1}{(s+\sqrt{2})(s+\sqrt{3})} = \frac{A}{s+\sqrt{2}} + \frac{B}{s+\sqrt{3}}$$

$$1 = A(s+\sqrt{3}) + B(s+\sqrt{2})$$

put $s = -\sqrt{3}$

$$1 = B(-\sqrt{3} + \sqrt{2})$$

$$\therefore B = \frac{1}{\sqrt{2} - \sqrt{3}}$$

$$\text{put } s = -\sqrt{2}$$

$$\therefore 1 = A(-\sqrt{2} + \sqrt{3})$$

$$A = \frac{1}{\sqrt{3} - \sqrt{2}}$$

$$\frac{1}{(s+\sqrt{2})(s+\sqrt{3})} = \frac{1}{(\sqrt{3}-\sqrt{2})(s+\sqrt{2})} + \frac{1}{(\sqrt{2}-\sqrt{3})(s+\sqrt{3})}$$

$$\frac{1}{s\sqrt{3} + \sqrt{6} - s\sqrt{2} - 2}$$

$$\frac{1}{s\sqrt{2} + \sqrt{6} - s\sqrt{3} - 3}$$

$$\frac{1}{s(\sqrt{3}-\sqrt{2})}$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{1}{(s+\sqrt{2})(s+\sqrt{3})}\right\} = \frac{1}{(\sqrt{3}-\sqrt{2})} \mathcal{L}^{-1}\left\{\frac{1}{s+\sqrt{2}}\right\} + \frac{1}{\sqrt{2}-\sqrt{3}} \mathcal{L}^{-1}\left\{\frac{1}{s+\sqrt{3}}\right\}$$

$$= \frac{1}{(\sqrt{3}-\sqrt{2})} \cdot e^{-\sqrt{2}t} + \frac{1}{(\sqrt{2}-\sqrt{3})} \cdot e^{-\sqrt{3}t}$$

=====

Q. $\mathcal{L}^{-1} \left\{ \frac{2}{s^4} - \frac{48}{s^6} \right\}$

Ans: $\mathcal{L}^{-1} \left\{ \frac{1}{s^4} \right\} - 48 \mathcal{L}^{-1} \left\{ \frac{1}{s^6} \right\}$

~~$= \frac{2 \cdot t^3}{s^3} - \frac{48 \cdot t^5}{s^5}$~~ $= \frac{2 \cdot t^3}{3!} - 48 \cdot \frac{t^5}{5!}$

Q. $\mathcal{L}^{-1} \left\{ \frac{4s+32}{s^2-16} \right\}$

Ans: $\mathcal{L}^{-1} \left\{ \frac{4s}{s^2-16} \right\} + \mathcal{L}^{-1} \left\{ \frac{32}{s^2-16} \right\}$

$= 4 \cosh 4t + 32 \frac{\sinh 4t}{4}$

Q. $\mathcal{L}^{-1} \left\{ \frac{1}{(s+a)(s+b)} \right\}$

$\frac{1}{(s+a)(s+b)} = \frac{A}{(s+a)} + \frac{B}{(s+b)}$

$\therefore 1 = A(s+b) + B(s+a)$

$\therefore s = -b$

$\therefore B = \frac{1}{a-b}$

$s = -a$

$A = \frac{1}{b-a}$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{1}{(s+a)(s+b)} \right\} = \frac{1}{(b-a)} e^{at} + \frac{1}{(a-b)} e^{bt}$$

Q. $\mathcal{L}^{-1} \left\{ \frac{1}{(s+2)^2} \right\}$

Ans: $\mathcal{L}^{-1} \left\{ \frac{1}{(s+2)^2} \right\}$

$s \rightarrow s+2$

$= e^{-2t} t$

$\frac{(t-2)'}{1!} e^{2t}$
 $= (t-2) e^{2t}$

Q. $\mathcal{L}^{-1} \left\{ \frac{(s-3)}{(s-3)^2 + 9} \right\}$

Ans: $s \rightarrow s-3 \quad \therefore e^{3t} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 3^2} \right\}$

$= e^{3t} \cos 3t$

Q. $\mathcal{L}^{-1} \left\{ \frac{-s+11}{s^2-2s-3} \right\}$

$= \mathcal{L}^{-1} \left\{ \frac{11-s}{(s-3)(s+1)} \right\}$

s^2-2s-3

$\therefore s=3 \text{ or } s=-1$

$$\therefore \frac{11-s}{(s-3)(s+1)} = \frac{A}{(s-3)} + \frac{B}{s+1}$$

$$\therefore 11-s = A(s+1) + B(s-3)$$

$$\text{let } s = -1$$

$$12 = B(-4)$$

$$\therefore \underline{\underline{B = -3}}$$

$$\text{let } s = 3$$

$$8 = A(4)$$

$$\therefore \underline{\underline{A = 2}}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{11-s}{(s-3)(s+1)} \right\} = A \mathcal{L}^{-1} \left\{ \frac{1}{s-3} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\}$$

$$= \underline{\underline{2e^{3t} - 3e^{-t}}}$$

Q. Find $\mathcal{L}^{-1} \left\{ \frac{4s+7}{s^2+2s+5} \right\}$.

(IF in calculator solution is in decimal not factorable)
Then use, square completion method.

steps : 1 Add and subtract ~~half the~~ $\left(\frac{1}{2} \text{ coe. of } s\right)^2$

$$\therefore s^2 + 2s + 5 = s^2 + 2s + 5 + 1 - 1$$

$$= (s^2 + 2s + 1) + 5 - 1$$

$$= \underline{\underline{(s+1)^2 + (2)^2}}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{4s+7}{s^2+2s+5} \right\} = \mathcal{L}^{-1} \left\{ \frac{4s+7}{(s+1)^2+2^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{4(s+1-1)+7}{(s+1)^2+2^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{4s+4-4+7}{(s+1)^2+2^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{4(s+1)+3}{(s+1)^2+2^2} \right\}$$

$$4 \mathcal{L}^{-1} \left\{ \frac{s+1}{(s+1)^2+2^2} \right\} + 3 \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2+2^2} \right\} = 4e^{-t} \cos 2t + \frac{3e^{-t} \sin 2t}{2}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{1}{s} + \frac{1}{s+2} + \frac{1}{s^2-4} \right\}$$

$$Ans. = \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s^2-4} \right\}$$

$$= 1 + e^{-2t} + \frac{\sinh 2t}{2}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{4s+12}{s^2+8s+16} \right\}$$

$s^2+8s+16$
 $s = -4, s = -4$

$$= \mathcal{L}^{-1} \left\{ \frac{4s+12}{(s+4)(s+4)} \right\}$$

$$= A \mathcal{L}^{-1} \left\{ \frac{1}{s+4} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s-1} \right\}$$

$$\therefore \frac{4s+12}{(s^2+4)(s-1)} = \frac{A}{s+4} + \frac{B}{s-1}$$

$$\therefore 4s+12 = A(s-1) + B(s+4)$$

$$\text{let } s=1$$

$$16 = B(5)$$

$$B = \frac{16}{5}$$

$$\text{let } s=-4$$

$$-4 = A(-5)$$

$$\therefore A = \frac{4}{5}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{4s+12}{s^2+4s+16} \right\} = \frac{4}{5} e^{-4t} + \frac{16}{5} e^t$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{1}{s^2-4s+8} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{(s-2)^2+4} \right\}$$

$$= \frac{e^{2t} \sin 2t}{2}$$

$$s^2-4s+8$$

$$= s^2-4s+8+4-4$$

$$= (s-2)^2+4$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{1}{s^2 - 5s + 6} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{(s-3)(s-2)} \right\}$$

$$= A \mathcal{L}^{-1} \left\{ \frac{1}{(s-3)} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{(s-2)} \right\}$$

$$1 = A(s-2) + B(s-3)$$

$$\therefore \text{let } s = 2$$

$$\therefore B = -1$$

$$\text{let } s = 3$$

$$A = 1$$

$$= \underline{\underline{e^{3t} - e^{2t}}}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{2\bar{u}}{(s+\bar{u})^3} \right\}$$

$$\underline{\underline{\frac{-1}{s}}}$$

$$= 2\bar{u} \mathcal{L}^{-1} \left\{ \frac{1}{(s+\bar{u})^3} \right\}$$

$$= \underline{\underline{2\bar{u} e^{-\bar{u}t} \cdot \frac{t^2}{2!}}}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{2\bar{u}}{3(s+\bar{u})} \right\}$$

$$= \frac{2\bar{u}}{3} \mathcal{L}^{-1} \left\{ \frac{1}{s+\bar{u}} \right\}$$

$$= \underline{\underline{\frac{2\bar{u}}{3} e^{-\bar{u}t}}}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{90}{(s+\sqrt{3})^6} \right\}$$

$$= 90 \mathcal{L}^{-1} \left\{ \frac{1}{(s+\sqrt{3})^6} \right\}$$

$$= 90 e^{-\sqrt{3}t} \frac{t^5}{5!}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{4}{s^2 - 2s - 3} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{4}{(s-3)(s+1)} \right\}$$

$$= A \mathcal{L}^{-1} \left\{ \frac{1}{(s-3)} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)} \right\}$$

$$4 = A(s+1) + B(s-3)$$

$$\text{let } s = -1$$

$$\therefore B = -1$$

$$\text{let } s = 3$$

$$A = 1$$

$$= \underline{\underline{e^{3t} - e^{-t}}}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{\bar{u}}{s^2 + 4s\pi + 3\pi^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{\pi}{(s+\pi)(s+3\pi)} \right\}$$

$$= A \mathcal{L}^{-1} \left\{ \frac{1}{(s+\pi)} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{(s+3\pi)} \right\}$$

$$\bar{u} = A(s+3\pi) + B(s+\pi)$$

$$\text{let } s = -3\pi$$

$$\therefore B = -1/2$$

$$\text{let } s = -\pi$$

$$\therefore A = 1/2$$

$$= \frac{1}{2} e^{-\pi t} - \frac{1}{2} e^{-3\pi t}$$

$$= \frac{1}{2} (e^{-\pi t} - e^{-3\pi t})$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{a_0}{s+1} + \frac{a_1}{(s+1)^2} + \frac{a_2}{(s+1)^3} \right\}$$

$$= a_0 \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} + a_1 \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2} \right\} + a_2 \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^3} \right\}$$

$$= a_0 e^{-t} + a_1 e^{-t} \frac{t}{1!} + a_2 e^{-t} \frac{t^2}{2!}$$

$$= e^{-t} \left(q_0 + \frac{a_1 t}{1} + \frac{a_2 t^2}{2} \right)$$

Q. Find $\mathcal{L}^{-1}$ of $\left\{ \frac{a(s+k) + b\pi}{(s+k)^2 + \pi^2} \right\}$

Ans: $= a \mathcal{L}^{-1} \left\{ \frac{s+k}{(s+k)^2 + \pi^2} \right\} + b\pi \mathcal{L}^{-1} \left\{ \frac{1}{(s+k)^2 + \pi^2} \right\}$

$$= a e^{-kt} \cos \pi t + e^{-kt} \frac{b\pi \sin \pi t}{\pi}$$

$$= e^{-kt} (a \cos \pi t + b \sin \pi t)$$

Q. Find $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)(s^2+9)} \right\}$ $\frac{s^4 + 10s^2 + 9}{(s^2+1)(s^2+9)}$

$$= \mathcal{L}^{-1} \left\{ \frac{As+B}{s^2+1} + \frac{Cs+D}{s^2+9} \right\}$$

~~Ans: 1~~

2. Find $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)(s^2+9)} \right\}$

put $s^2 = p$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{1}{(p+1)(p+9)} \right\}$$

$$\therefore \frac{1}{(p+1)(p+9)} = \frac{A}{(p+1)} + \frac{B}{p+9}$$

$$1 = A(p+9) + B(p+1)$$

let $p = -9$

$$\therefore B = \frac{-1}{8}$$

let $p = -1$

$$\therefore A = \frac{1}{8}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{1}{(p+1)(p+9)} \right\} = \frac{1}{8} e^{-t} - \frac{1}{8} e^{-3t}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1/8}{(p+1)} + \frac{-1/8}{(p+9)} \right\}$$

$$= \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{1}{s^2+1} \right\} - \frac{1}{8} \mathcal{L}^{-1} \left\{ \frac{1}{s^2+9} \right\}$$

$$= \frac{1}{8} \sin t - \frac{1}{8} \frac{\sin 3t}{3}$$

Differentiation of Transforms (Multiplication by t)

If $L\{f(t)\} = F(s)$ then ;

$$L\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$$

proof :

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

$$\frac{d}{ds} F(s) = \int_0^{\infty} e^{-st} \cdot (-t) f(t) dt \quad \left[\begin{array}{l} \text{w.r.to 's', } t \text{ is a} \\ \text{const} \end{array} \right]$$

$$-\frac{d}{ds} F(s) = \int_0^{\infty} t \cdot e^{-st} f(t) dt$$

$$-\frac{d}{ds} F(s) = L\{t f(t)\}$$

similarly,

$$L\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$$

$$L\{t^2 \cdot f(t)\} = \frac{d^2}{ds^2} F(s)$$

$$L\{t^3 \cdot f(t)\} = -\frac{d^3}{ds^3} F(s)$$

$$L\{t^4 \cdot f(t)\} = \frac{d^4}{ds^4} F(s)$$

$$\vdots$$
$$L\{t^n \cdot f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$$

Q. Find $L\{t \cdot \sin \omega t\}$

Ans: $L\{t \cdot \sin \omega t\} = (-1)' \frac{d}{ds} \left[\int_0^\infty e^{-st} \sin \omega t dt \right]$

$$= (-1)' \cdot \frac{d}{ds} \left[\int_0^\infty e^{-st} \sin \omega t dt \right]$$

$$= -1 \int_0^\infty e^{-st} \cdot -s \sin \omega t dt$$

$$= \int_0^\infty s e^{-st} \sin \omega t dt$$

Q. Find $L\{t \cdot \sin \omega t\}$

Ans: $L\{t \cdot \sin \omega t\} = (-1)' \cdot \frac{d}{ds} L\{\sin \omega t\}$

$$= -1 \cdot \frac{d}{ds} \left(\frac{\omega}{s^2 + \omega^2} \right)$$

$$= -\omega \frac{d}{ds} \left(\frac{1}{s^2 + \omega^2} \right)$$

$$= -\omega \cdot \left(\frac{-1}{(s^2 + \omega^2)^2} \cdot 2s \right)$$

$$= \frac{2s\omega}{(s^2 + \omega^2)^2}$$

Q. find $L\{t^3 \cdot e^{-3t}\}$

$$\text{Ans: } L\{t^3 \cdot e^{-3t}\} = \frac{3!}{(s+3)^4}$$

Q. find $L\{t^2 \cdot \sinh 2t\}$

$$\text{Ans: } L\{t^2 \cdot \sinh 2t\} = (-1)^2 \frac{d^2}{ds^2} L\{\sinh 2t\}$$

$$= \frac{d^2}{ds^2} \left(\frac{2}{s^2 - 2^2} \right)$$

$$= 2 \cdot \frac{d}{ds} \left(\frac{-1}{(s^2 - 2^2)^2} \cdot 2s \right)$$

$$= -4 \frac{d}{ds} \frac{s}{(s^2 - 2^2)^2}$$

$$= -4 \left[\frac{(s^2 - 4)^2 - (s) 2(s^2 - 4) \cdot 2s}{(s^2 - 4)^4} \right]$$

$$= -4 \left[\frac{(s^2 - 4) \cdot ((s^2 - 4) - 4s^2)}{(s^2 - 4)^4} \right]$$

$$= -4 \frac{[(s^2 - 4) - 4s^2]}{(s^2 - 4)^3}$$

$$= \frac{-4s^2 + 16 + 16s^2}{(s^2 - 4)^3} = \frac{12s^2 + 16}{(s^2 - 4)^3}$$

OR

$$\mathcal{L}\left\{t^2 \left(\frac{e^{2t} - e^{-2t}}{2}\right)\right\}$$

$$= \frac{1}{2} \left[\mathcal{L}\{t^2 e^{2t}\} - \mathcal{L}\{t^2 e^{-2t}\} \right]$$

$$= \frac{1}{2} \left[(-1)^2 \frac{d^2}{ds^2} \mathcal{L}\{e^{2t}\} - (-1)^2 \frac{d^2}{ds^2} \mathcal{L}\{e^{-2t}\} \right]$$

$$= \frac{1}{2} \left[-\frac{2!}{(s+2)^3} + \frac{2!}{(s-2)^3} \right]$$

Q. Find $\mathcal{L}\{t \cdot e^{-t} \cdot \cos t\}$

Consider

$$\mathcal{L}\{t \cdot \cos t\}$$

$$= (-1) \frac{d}{ds} \mathcal{L}\{\cos t\}$$

$$= - \frac{d}{ds} \frac{s}{s^2 + 1}$$

$$= - \left[\frac{(s^2 + 1) - (s)(2s)}{(s^2 + 1)^2} \right]$$

$$= \frac{s^2 - 1}{(s^2 + 1)^2}$$

when $s \rightarrow s+1$

$$\mathcal{L}\{t \cdot e^{-t} \cdot \cos t\} = \frac{(s+1)^2 - 1}{(s+1)^2 + 1}$$

$$Q. \mathcal{L}\{t \cdot e^{-kt} \cdot \sin t\}$$

$$\text{consider : } \mathcal{L}\{t \cdot \sin t\}$$

$$= (-1)^1 \cdot \frac{d}{ds} \mathcal{L}\{\sin t\}$$

$$= -\frac{d}{ds} \frac{1}{s^2+1}$$

$$= -\left[\frac{-2s}{(s^2+1)^2} \right]$$

$$= \frac{2s}{(s^2+1)^2}$$

$$\text{when } s \rightarrow s+k$$

$$= \frac{2(s+k)}{((s+k)^2+1)^2}$$

$$Q. \mathcal{L}\{t \cdot e^{2t} \cdot \sin 3t\}$$

$$\text{consider } \mathcal{L}\{t \cdot \sin 3t\}$$

$$= (-1)^1 \cdot \frac{d}{ds} \mathcal{L}\{\sin 3t\}$$

$$= -\frac{d}{ds} \frac{3}{s^2+9}$$

$$= -3 \cdot \frac{-2s}{(s^2+9)^2}$$

$$= \frac{6s}{(s^2+9)^2}$$

$$s \rightarrow s-2$$

$$= \frac{6(s-2)}{\underline{\underline{((s-2)^2+9)^2}}}$$

Q. $L\{3t \cdot \sinh 4t\}$

$$= 3 L\{t \cdot \sinh 4t\}$$

$$= 3 \cdot (-1)^1 \cdot \frac{d}{ds} L\{\sinh 4t\}$$

$$= -3 \frac{d}{ds} \frac{4^2}{s^2 - 4^2}$$

$$= -12 \cdot \frac{-2s}{(s^2 - 4^2)^2}$$

$$= \frac{24s}{(s^2 - 4^2)^2}$$

Q. $L\left\{\frac{t^2}{2} \cos \frac{\pi}{2} t\right\}$

$$= \frac{1}{2} L\left\{t^2 \cdot \cos \frac{\pi}{2} t\right\}$$

$$= \frac{1}{2} \cdot \frac{2b - (p+s)^2}{(p+s)^2}$$

$$= \frac{1}{2} \cdot (-1)^2 \cdot \frac{d^2}{ds^2} \left\{ \cos \frac{\pi t}{2} \right\}$$

$$= \frac{1}{2} \frac{d^2}{ds^2} \left(\frac{s}{s^2 + \left(\frac{\pi}{2}\right)^2} \right)$$

$$= \frac{1}{2} \left[\frac{(s^2 + \frac{\pi^2}{4}) \cdot (-2s) - (s) \cdot 2s}{(s^2 + \frac{\pi^2}{4})^2} \right] \quad \left(\frac{\pi^2}{4} - s^2 \right)$$

$$= \frac{1}{2} \left[\frac{(s^2 + \frac{\pi^2}{4})(-2s) - 2(s^2 + \frac{\pi^2}{4})s \cdot \left(\frac{\pi^2}{4} - s^2\right)}{(s^2 + \frac{\pi^2}{4})^4} \right]$$

$$= \frac{1}{2} \left[\frac{-2s^3 - \frac{\pi^2 s}{2}}{(s^2 + \frac{\pi^2}{4})^4} \right]$$

$$= \frac{1}{2} \left[\frac{(s^2 + \frac{\pi^2}{4}) \left[-2s^3 - \frac{\pi^2 s}{2} + 4s^3 - \pi^2 s \right]}{(s^2 + \frac{\pi^2}{4})^4} \right]$$

$$= \frac{1}{2} \left[\frac{2s^3 - \frac{3}{2} \pi^2 s}{(s^2 + \frac{\pi^2}{4})^3} \right]$$

$$= \frac{16s(4s^2 - 3\pi^2)}{(4s^2 + \pi^2)^3}$$

Inverse

proof:

$$\text{If } \mathcal{L}\{t f(t)\} = -\frac{d}{ds} F(s)$$

multiplying by $\mathcal{L}^{-1}$ in both sides;

$$\mathcal{L}^{-1} = I$$

$$\therefore t f(t) = \mathcal{L}^{-1}\left\{-\frac{d}{ds} F(s)\right\}$$

$$f(t) = \frac{1}{t} \mathcal{L}^{-1}\left\{-\frac{d}{ds} F(s)\right\}$$

Q Find $\mathcal{L}^{-1}\left\{\log \frac{(s+a)}{(s-a)}\right\}$

Ans: STEP 1: $F(s) = \log \frac{(s+a)}{(s-a)}$

STEP 2: $\frac{d}{ds} F(s) = \frac{d}{ds} [\log(s+a) - \log(s-a)]$

$$\frac{d}{ds} F(s) = \frac{1}{s+a} - \frac{1}{s-a}$$

$$-\frac{d}{ds} F(s) = \frac{1}{s-a} - \frac{1}{s+a}$$

STEP 3: $\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$

$$\therefore \mathcal{L}\{t \cdot f(t)\} = \frac{1}{s-a} - \frac{1}{s+a}$$

$$\therefore t \cdot f(t) = \mathcal{L}^{-1}\left\{\frac{1}{s-a} - \frac{1}{s+a}\right\}$$

STEP 4) $\therefore t \cdot f(t) = e^{at} - e^{-at}$

$$f(t) = \frac{1}{t} (e^{at} - e^{-at})$$

STEP 5:) $\mathcal{L}^{-1} \left\{ \log \left(\frac{s+a}{s-a} \right) \right\} = \mathcal{L}^{-1} \{ F(s) \} = f(t)$

$$= \frac{e^{at} - e^{-at}}{t}$$

$$= \frac{2 \sinh at}{t}$$

Q. Find $\mathcal{L}^{-1} \left\{ \cot^{-1} \left(\frac{s}{a} \right) \right\}$

$$F(s) = \cot^{-1} \left(\frac{s}{a} \right)$$

$$-\frac{d}{ds} F(s) = - \left[\frac{-1}{1 + \left(\frac{s}{a} \right)^2} \right] \times \frac{1}{a} = \frac{1}{a(1 + \frac{s^2}{a^2})}$$

$$-\frac{d}{ds} F(s) = \frac{1}{a \left(\frac{a^2 + s^2}{a^2} \right)} = \frac{a}{a^2 + s^2}$$

$$-\frac{d}{ds} F(s) = \frac{a}{a^2 + s^2}$$

$$\therefore \mathcal{L} \{ t \cdot f(t) \} = -\frac{d}{ds} F(s)$$

$$t \cdot f(t) = \mathcal{L}^{-1} \left\{ \frac{a}{s^2 + a^2} \right\}$$

$$f(t) = \frac{1}{t} \cdot \sin at$$

$$\therefore \mathcal{L}^{-1}\left\{\cot^{-1}\frac{s}{a}\right\} = \frac{\sin at}{t}$$

Q. $\mathcal{L}^{-1}\left\{\tan^{-1}\left(\frac{2}{s}\right)\right\}$

Ans. $F(s) = \tan^{-1}\left(\frac{2}{s}\right)$

$$-\frac{d}{ds}F(s) = -\left[\frac{1}{1+\frac{4}{s^2}} \cdot \frac{-2}{s^2}\right]$$

$$= \frac{2}{\frac{s^2+4}{s^2}} \cdot s^x$$

$$\frac{2}{s^2+4}$$

$$\therefore \mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds}\left[F(s) \cdot \frac{1}{(s/2)+1}\right] = (2) \cdot \frac{2}{10}$$

$$t \cdot f(t) = \mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} \cdot \frac{1}{s/2+1} = (2) \cdot \frac{2}{10}$$

$$f(t) = \frac{1}{t}(\sin 2t)$$

$$\therefore \mathcal{L}^{-1}\left\{\tan^{-1}\left(\frac{2}{s}\right)\right\} = \frac{\sin 2t}{t}$$

$$Q. \mathcal{L}^{-1} \left\{ \ln \left(1 + \frac{\omega^2}{s^2} \right) \right\}$$

$$\log ab = \log a + \log b$$

$$\log \frac{a}{b} = \log a - \log b$$

$$\text{Ans. } F(s) = \ln \left(1 + \frac{\omega^2}{s^2} \right)$$

$$\begin{aligned} \text{step 1)} \quad &= \ln \left(\frac{s^2 + \omega^2}{s^2} \right) \\ &= \ln(s^2 + \omega^2) - \ln(s^2) \end{aligned}$$

$$\text{step 2)} \quad \frac{d}{ds} F(s) = - \left[\frac{2s}{s^2 + \omega^2} - \frac{2s}{s^2} \right]$$

$$\frac{d}{ds} F(s) = \left(\frac{2}{s} - \frac{2s}{s^2 + \omega^2} \right)$$

$$\text{step 3)} \quad \mathcal{L}\{t \cdot f(t)\} = - \frac{d}{ds} F(s)$$

$$t \cdot f(t) = \mathcal{L}^{-1} \left\{ \frac{2}{s} - \frac{2s}{s^2 + \omega^2} \right\}$$

$$\text{step 4)} \quad f(t) = \frac{1}{t} (2 - 2 \cos \omega t)$$

$$\text{step 5)} \quad \mathcal{L}^{-1} \left\{ \ln \left(1 + \frac{\omega^2}{s^2} \right) \right\} = \frac{2(1 - \cos \omega t)}{t}$$

$$Q. \mathcal{L}^{-1} \left\{ \ln \left(\frac{s}{s-1} \right) \right\}$$

$$\text{Ans. } F(s) = \ln \left(\frac{s}{s-1} \right)$$

$$= \ln(s-1) - \ln(s)$$

$$\frac{d}{ds} F(s) = \frac{1}{s-1} - \frac{1}{s}$$

$$-\frac{d}{ds} F(s) = \frac{1}{s-1} - \frac{1}{s-2}$$

$$\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$$

$$t \cdot f(t) = \mathcal{L}^{-1}\left\{-\left(\frac{1}{s-1} - \frac{1}{s-2}\right)\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-1} - \frac{1}{s-2}\right\}$$

$$f(t) = \frac{1}{t} (e^t - e^{2t}) = \frac{1}{t} (e^t - 1)$$

Ar, exams
tan⁻¹
cot⁻¹
log, log

Q. $\mathcal{L}^{-1}\left\{\text{arc cot}\left(\frac{s}{a}\right)\right\}$

Ans. $\text{arc cot}\left(\frac{s}{a}\right) = \cot^{-1}\left(\frac{s}{a}\right)$

$$= \mathcal{L}^{-1}\left\{\cot^{-1}\left(\frac{s}{a}\right)\right\}$$

$$F(s) = \cot^{-1}\left(\frac{s}{a}\right)$$

$$\frac{d}{ds} F(s) = \frac{-1}{1 + \frac{s^2}{a^2}} \times \frac{1}{a} = \frac{-a^2}{s^2 + a^2} \times \frac{1}{a}$$

$$-\frac{d}{ds} F(s) = \frac{\pi^2}{s^2 + a^2}$$

$$\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds} F(s) =$$

$$f(t) = \frac{1}{t} \mathcal{L}^{-1}\left\{\frac{\pi^2}{s^2 + a^2}\right\}$$

$$= \frac{1}{t} \sin at$$

~~$$L\left\{\ln \frac{s^2+1}{(s-1)^2}\right\}$$~~

Division by 't'

If $L\{f(t)\} = F(s)$ then $L\left\{\frac{f(t)}{t}\right\}$ is ;

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s) ds$$

proof :

$$F(s) = \int_0^\infty e^{-st} f(t) dt$$

$$\int_s^\infty F(s) ds = \int_s^\infty \int_0^\infty e^{-st} f(t) dt$$

$$= \int_0^\infty \int_s^\infty e^{-st} ds \cdot f(t) dt$$

$$= \int_0^\infty \left[\frac{e^{-st}}{-t} \right]_{s,s}^\infty f(t) dt$$

$$= \int_0^\infty \frac{e^{-st}}{t} f(t) dt$$

$$= \int_0^\infty e^{-st} \frac{f(t)}{t} dt$$

$$= L\left\{\frac{f(t)}{t}\right\}$$

Q. $L \left\{ \frac{\cos 2t - \cos 3t}{t} \right\}$

Ans.

$$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty \frac{f(s)}{s} ds = \int_s^\infty f(s) ds$$

$$= L \left\{ \frac{\cos 2t}{t} \right\} - L \left\{ \frac{\cos 3t}{t} \right\}$$

$$= \int_s^\infty \cos 2t dt - \int_s^\infty \cos 3t dt$$

$$= \left[\frac{\sin 2t}{2} \right]_s^\infty - \left[\frac{\sin 3t}{3} \right]_s^\infty$$

$$= \frac{\sin 2s}{2} - \frac{\sin 3s}{3}$$

Q. $L \left\{ \frac{\cos 2t - \cos 3t}{t} \right\}$

~~$F(s) = \cos 2t - \cos 3t$~~

Q. $L \left\{ \frac{\cos 2t - \cos 3t}{t} \right\}$

$F(s) = L \{ \cos 2t - \cos 3t \} = \frac{s}{s^2 + 2^2} - \frac{s}{s^2 + 3^2}$

$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty f(s) ds$

$= \int_s^\infty \frac{s}{s^2 + 4} - \frac{s}{s^2 + 9} ds$

$= \left[\frac{1}{2} \ln(s^2 + 4) - \frac{1}{2} \ln(s^2 + 9) \right]_{s=s}^{s=\infty}$

$$= \frac{1}{2} \ln \left(\frac{s^2+4}{s^2+9} \right) \Big|_s^\infty$$

(log(∞) = not defined)

$$= \frac{1}{2} \ln \left(\frac{s^2 \left(1 + \frac{4}{s^2} \right)}{s^2 \left(1 + \frac{9}{s^2} \right)} \right) \Big|_s^\infty$$

$$= \frac{1}{2} \left[\ln(1) - \log \left(\frac{s^2+4}{s^2+9} \right) \right] = -\frac{1}{2} \log \left(\frac{s^2+4}{s^2+9} \right)$$

Defn $= \frac{1}{2} \log \left(\frac{s^2+9}{s^2+4} \right)$

Q. find $L \left\{ \frac{e^{-at} - e^{-bt}}{t} \right\}$

$$= \cancel{L \{ \frac{e^{-at}}{t} \} - L \{ \frac{e^{-bt}}{t} \}}$$

$$= L \left\{ \frac{e^{-at}}{t} \right\} - L \left\{ \frac{e^{-bt}}{t} \right\}$$

$$= \int_s^\infty \frac{1}{s+a} - \int_s^\infty \frac{1}{s+b}$$

$$= \left[\ln(s+a) - \ln(s+b) \right]_s^\infty$$

$$= \left[\ln \left(\frac{s+a}{s+b} \right) \right]_s^\infty$$

$$= \left[\ln \left(\frac{1 + \frac{a}{s}}{1 + \frac{b}{s}} \right) \right]_s^\infty$$

$$= \log(1) - \log \left(\frac{s+a}{s+b} \right) = \log \left(\frac{s+b}{s+a} \right)$$

Q. $L \left\{ \frac{\sin at}{t} \right\}$

Ans. $= \int_0^\infty L \{ \sin at \}$

$= \int_0^\infty \frac{2}{s^2 + a^2} ds$

$= 2 \int_0^\infty \frac{1}{s^2 + a^2} ds$

$= \left[\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \right]$

$= 2 \cdot \left[\frac{1}{a} \tan^{-1} \left(\frac{s}{a} \right) \right]_0^\infty$

$= \left[\tan^{-1} \left(\frac{s}{a} \right) \right]_0^\infty$

$= \tan^{-1}(\infty) - \tan^{-1} \left(\frac{0}{a} \right)$
 $= \frac{\pi}{2} - \tan^{-1} \left(\frac{0}{a} \right)$
 $= \frac{\pi}{2}$

Q. $L \left\{ \frac{1 - \cos at}{t} \right\}$

Ans. $= \int_0^\infty L \{ 1 - \cos at \} ds$

$= \int_0^\infty \left(\frac{1}{s^2} - \frac{1}{s^2 + a^2} \right) ds$

$\left[\left(\frac{p+1}{p+2} \right) \right] \text{ at } \frac{1}{s} =$

$\left[\left(\frac{\frac{p}{2}+1}{\frac{p}{2}+2} \right) \right] \text{ at } \frac{1}{s} =$

$\left[\left(\frac{\frac{p+1}{2}}{\frac{p+2}{2}} \right) \right] \text{ at } \frac{1}{s} =$

$\left[\left(\frac{1}{2} \right) \right] \text{ at } \frac{1}{s} =$

$\left[\left(\frac{1}{2} \right) \right] \text{ at } \frac{1}{s} =$

$\left[\left(\frac{1}{2} \right) \right] \text{ at } \frac{1}{s} =$

$\left[\left(\frac{1}{2} \right) \right] \text{ at } \frac{1}{s} =$

$$\begin{aligned}
 &= \int_s^\infty \left[\frac{1}{s} - \frac{s}{s^2+1} \right] \\
 &= \left[\log(s) - \frac{1}{2} \log(1+s^2) \right]_s^\infty \\
 &= \left[\log\left(\frac{\cancel{1+s^2}}{s}\right) \right]_s^\infty = \left[\log\left(\frac{s}{\sqrt{s^2+1}}\right) \right]_s^\infty \\
 &= \left[\log\left(\frac{s \sqrt{\frac{1}{s^2+1}}}{s}\right) \right]_s^\infty = \log\left(\frac{1}{\sqrt{1+\frac{1}{s^2}}}\right)_s^\infty \\
 &= \log(1) - \log \frac{\sqrt{1+s^2}}{s} = \log 1 - \log \frac{s}{\sqrt{s^2+1}} \\
 &= \log\left(\frac{s}{\sqrt{1+s^2}}\right)
 \end{aligned}$$

a. $L \left\{ \left(\frac{\sin 2t}{\sqrt{t}} \right)^2 \right\}$

$$= L \left\{ \frac{(\sin 2t)^2}{t} \right\}$$

$$= \int_s^\infty L \{ (\sin 2t)^2 \} ds$$

$$= \int_s^\infty L \left\{ \frac{1 - \cos 4t}{2} \right\} ds$$

$$= \frac{1}{2} \int_s^\infty \left[\frac{1}{s} - \frac{s}{s^2+4^2} \right] ds$$

$$\begin{aligned}
 &= \frac{1}{4} \left[\log(s) - \frac{1}{2} \log(s^2 + 4^2) \right]_s^\infty \\
 &= \frac{1}{4} \left[\log \frac{s}{\sqrt{s^2 + 4^2}} \right]_s^\infty \\
 &= \frac{1}{4} \left[\log \frac{1}{\sqrt{1 + \frac{4^2}{s^2}}} \right]_s^\infty \\
 &= \frac{1}{4} \left[\log(1) - \log \frac{s}{\sqrt{s^2 + 4^2}} \right] \\
 &= \frac{1}{4} \log \frac{\sqrt{s^2 + 4^2}}{s}
 \end{aligned}$$

Laplace Transform of Integrals.

If $L\{f(t)\} = F(s)$ then,

$$L\left\{\int_0^t f(t) dt\right\} = \frac{F(s)}{s}$$

Similarly $L\left\{\int_0^t \int_0^t f(t) dt dt\right\} = \frac{F(s)}{s^2}$

Q. Find $L\left\{\int_0^t \cos t dt\right\}$

$$= \frac{1}{s} [L\{\cos t\}]$$

$$= \left(\frac{s}{s^2 + 1}\right) \times \frac{1}{s}$$

$$= \frac{1}{s^2 + 1}$$

~~VARIABLE~~

Q. Find $L \{ e^{-4t} \int_0^t t \cdot \sin t \, dt \}$

Ans: Consider

$$L \{ \int_0^t t \cdot \sin t \, dt \}$$

$$= \frac{1}{s} L \{ \sin t \}$$

$$= \frac{1}{(s^2 + \omega^2)s}$$

when $s \rightarrow s+4$

$$= \frac{1}{((s+4)^2 + \omega^2)s+4}$$

Q. Find $L \{ e^{-4t} \int_0^t t \cdot \sin t \, dt \}$

Consider $L \{ \int_0^t t \cdot \sin t \, dt \}$

$$= \frac{1}{s} L \{ t \cdot \sin t \}$$

$$= \frac{1}{s} \cdot -\frac{d}{ds} L \{ \sin t \}$$

$$= -\frac{1}{s} \cdot \frac{d}{ds} \left[\frac{1}{s^2 + 1} \right]$$

$$= -\frac{1}{s} \cdot \frac{-2s}{(s^2 + 1)^2} = \frac{2}{s(s^2 + 1)^2}$$

$s \rightarrow s+4$

$$= \frac{2}{(s+4)^2 + 1)^2}$$

Q. $L\{e^{-st} \int_0^t \frac{\sin t}{t} dt\}$

Consider, $L\left\{\int_0^t \frac{\sin t}{t} dt\right\}$

$$= \frac{1}{s} L\left\{\frac{\sin t}{t}\right\}$$

$$= \frac{1}{s} \int_0^\infty L\{\sin t\} dt$$

$$= \frac{1}{s} \int_0^\infty \frac{1}{s^2+1} dt$$

$$= \frac{1}{s} [\tan^{-1} s]_0^\infty$$

$$= \frac{1}{s} \left[\frac{\pi}{2} - \tan^{-1} s \right]$$

when $s \rightarrow s+3$

$$\Rightarrow \frac{1}{(s+3)} \left[\frac{\pi}{2} - \tan^{-1}(s+3) \right]$$

INVERSE

we have,

$$L\left\{\int_0^t f(t) dt\right\} = \frac{F(s)}{s}$$

where $F(s) = L\{f(t)\}$

Apply L^{-1} on both sides,

$$\int_0^t f(t) dt = L^{-1}\left\{\frac{F(s)}{s}\right\}$$

$$\therefore L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(t) dt$$

similarly $\mathcal{L}^{-1}\left\{\frac{F(s)}{s^2}\right\} = \int_0^t \int_0^t f(t) dt$

2. Find $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+\omega^2)}\right\}$

$$= \int_0^t \frac{\sin \omega t}{\omega} dt$$

$$= \frac{1}{\omega} \int_0^t \sin \omega t dt$$

$$= \frac{1}{\omega} \left[\frac{-\cos \omega t}{\omega} \right]_0^t$$

$$= \frac{-1}{\omega^2} (\cos \omega t - \cos 0)$$

$$= \frac{1}{\omega^2} (1 - \cos \omega t)$$

Q. $\mathcal{L}^{-1}\left\{\frac{s+8}{s^2+4s^2}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2} \cdot \frac{s+8}{s^2+4}\right\}$$

$$\frac{1}{s^2} \cdot \frac{s+8}{s^2+4} = \frac{Ax+B}{s^2} + \frac{Cx+D}{s^2+4}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2} \cdot \frac{(s+8)}{(s+8)^2+4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s^2} \cdot \frac{(s+8)}{(s+8)^2+4}\right\}$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{s+8}{s^2+4s^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s^2} \cdot \frac{(s+8)}{s^2+4} \right\}$$

$$= \int_0^t \int_0^t \mathcal{L}^{-1} \left\{ \frac{(s+8)}{(s^2+4)} \right\} dt$$

$$= \int_0^t \int_0^t \mathcal{L}^{-1} \left\{ \frac{s}{s^2+4} \right\} + \mathcal{L}^{-1} \left\{ \frac{8}{s^2+4} \right\} dt$$

$$= \int_0^t \int_0^t \cos 2t + 4 \sin 2t dt$$

$$= \int_0^t \left[\frac{\sin 2t}{2} - \frac{4 \cos 2t}{2} \right]_0^t dt$$

$$= \int_0^t \frac{1}{2} \left[\sin 2t - 4 \cos 2t + 4 \right] dt$$

$$= \frac{1}{2} \left[-\frac{\cos 2t}{2} - \frac{4 \sin 2t}{2} + 4t \right]_0^t$$

$$= \frac{1}{2} \left[-\frac{\cos 2t}{2} - \frac{4 \sin 2t}{2} + 4t + \frac{1}{2} \right]$$

$$= \sin 2t - \frac{\cos 2t}{4} + 2t + \frac{1}{4}$$

$$Q \quad \mathcal{L}^{-1} \left\{ \frac{1}{s^4 - s^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s^2(s^2 - 1)} \right\}$$

$$= \int_0^t \int_0^t \mathcal{L}^{-1} \left\{ \frac{1}{s^2 - 1} \right\} dt dt$$

$$= \int_0^t \int_0^t \sinh t dt dt$$

$$= \int_0^t [-\cosh t]$$

$$= \int_0^t \int_0^t$$

$$= \int_0^t (\cosh t) dt$$

$$= \int_0^t \cosh t - 1 dt$$

$$= [\sinh t - t]_0^t$$

$$= \underline{\underline{\sinh t - t}}$$

$$\frac{d \sinh x}{dx} = \cosh x$$

$$\frac{d \cosh x}{dx} = \sinh x$$

$$\int \sinh x = \cosh x$$

$$\int \cosh x = \sinh x$$

$$\sinh 0 = 0$$

$$\cosh 0 = 1$$

Convolution

If $f(t)$ and $g(t)$ are two functions then their convolution is denoted by,

$$f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

Convolution Theorem

(given)

$$\text{If } \mathcal{L}\{f(t)\} = F(s)$$

$$\mathcal{L}\{g(t)\} = G(s)$$

then,

$$\mathcal{L}\{f(t) * g(t)\} = F(s) \cdot G(s)$$

ie ;

$$\mathcal{L}^{-1}[F(s) \cdot G(s)] = f(t) * g(t)$$

$$= \int_0^t f(u) \cdot g(t-u) du$$

Q. Find $1 * (-1)$

Ans:

$$\cancel{f(t) * g(t)}^t$$

$$f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

$$f(t) = 1$$

$$f(u) = 1$$

$$g(t) = -1$$

$$g(t-u) = -1$$

$$\begin{aligned} \therefore 1 * (-1) &= \int_0^t 1 \cdot (-1) du \\ &= -[u]_0^t \\ &= -t \end{aligned}$$

Q- Find $\cos wt * 1$

$$f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

$$f(t) = \cos wt \quad f(u) = \cos wu$$

$$g(t) = 1 \quad g(t-u) = 1$$

$$\cos wt * 1 = \int_0^t \cos wu \cdot 1 du$$

$$= \int_0^t \cos wu du$$

$$= \left[\frac{\sin wu}{w} \right]_0^t$$

$$= \frac{\sin wt}{w}$$

Q. Find $L^{-1} \left\{ \frac{1}{s(s-a)} \right\}$ using convolution theorem.

Ans: $L^{-1} \left\{ \frac{1}{(s-a)s} \right\}$ [Take the 2nd function as a simpler function]

$$L^{-1} \left\{ \frac{1}{(s-a)} \cdot \frac{1}{s} \right\}$$

$$f(s) = \frac{1}{s-a}$$

$$f(t) = e^{at}$$

$$f(u) = e^{au}$$

$$g(s) = \frac{1}{s}$$

$$g(t) = 1$$

$$g(t-u) = 1$$

$$f(t) * g(t-u) = 1$$

$$\begin{aligned}
 \mathcal{L}^{-1}\{F(s) \cdot G(s)\} &= f(t) * g(t) \\
 &= \int_0^t f(u) \cdot g(t-u) du \\
 &= \int_0^t e^{au} \cdot 1 du \\
 &= \left[\frac{e^{au}}{a} \right]_0^t \\
 &= \frac{1}{a} (e^{at} - 1)
 \end{aligned}$$

Q. using convolution theorem find $\mathcal{L}^{-1}\left\{\frac{40.5}{s(s^2-9)}\right\}$

Ans. $40.5 \mathcal{L}^{-1}\left\{\frac{1}{s(s^2-9)}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2-9)} \cdot \frac{1}{s}\right\}$$

$$F(s) = \frac{1}{s^2-9}$$

$$f(t) = \frac{\sinh 3t}{3}$$

$$f(u) = \frac{\sinh 3u}{3}$$

$$G(s) = \frac{1}{s}$$

$$g(t) = 1$$

$$g(t-u) = 1$$

$$\mathcal{L}^{-1}\{F(s) \cdot G(s)\} = f(t) * g(t)$$

$$= \int_0^t f(u) \cdot g(t-u) du$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{40.5}{s(s^2-9)}\right\} = \left[\int_0^t \frac{\sinh 3u}{3} \cdot 1 du \right] \times 40.5$$

$$= \frac{40.5}{3} \times \left[\frac{\cos 3ht}{3} \right]_0^t$$

$$= \frac{40.5}{3} (\cos 3ht - 1)$$

$$Q. \mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \omega^2)^2} \right\}$$

$$\text{Ans: } \mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \omega^2)(s^2 + \omega^2)} \right\}$$

$$\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

$$F(s) = \frac{1}{s^2 + \omega^2}$$

$$f(t) = \frac{\sin \omega t}{\omega}$$

$$f(u) = \frac{\sin \omega u}{\omega}$$

$$G(s) = \frac{1}{s^2 + \omega^2}$$

$$g(t) = \frac{\sin \omega t}{\omega}$$

$$g(t-u) = \frac{\sin \omega(t-u)}{\omega}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + \omega^2)^2} \right\} = \int_0^t \frac{\sin \omega u}{\omega} \cdot \frac{\sin \omega(t-u)}{\omega} du$$

$$= \frac{1}{\omega^2} \int_0^t \sin \omega u \cdot \sin \omega(t-u) du$$

$$= \frac{1}{2\omega^2} \int_0^t \cos(\omega u - \omega t + \omega u) - \cos(\omega u + \omega t - \omega u) du$$

$$= \frac{1}{2\omega^2} \int_0^t \cos(2\omega u - \omega t) - \cos(\omega t) du$$

$$= \frac{1}{2\omega^2} \left[\frac{\sin(2\omega u - \omega t)}{2\omega} - \cos(\omega t) \cdot u \right]_0^t$$

$$= \frac{1}{2\omega^2} \sin \omega(2L - t)$$

$$= \frac{1}{2\omega^2} \left[\frac{\sin \omega t}{2\omega} - t \cos \omega t \right] - \frac{\sin(\omega t)}{2\omega}$$

$$= \frac{1}{2\omega^2} \left[\frac{\sin(\omega t)}{2\omega} + \frac{\sin(\omega t)}{2\omega} - t \cos \omega t \right]$$

$$= \frac{1}{2\omega^2} \left[\frac{\sin \omega t}{\omega} - t \cos \omega t \right]$$

Q Find $\mathcal{L}^{-1} \left\{ \frac{\omega}{s^2(s^2 - \omega^2)} \right\}$

$$\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t F(u) \cdot g(t-u) du$$

$$F(s) = \frac{1}{s^2 - \omega^2} \quad f(t) = \frac{\sinh \omega t}{\omega} \quad f(u) = \frac{\sinh \omega u}{\omega}$$

$$G(s) = \frac{1}{s^2} \quad g(t) = \frac{t}{1!} \quad g(t-u) = t-u$$

$$\mathcal{L}^{-1} \left\{ \frac{\omega}{s^2(s^2 - \omega^2)} \right\} = \int_0^t \frac{\sinh \omega u}{\omega} \cdot (t-u) du$$

$$= \int_0^t t \sinh \omega u - u \sinh \omega u du$$

$$= \frac{1}{\omega} \left[\frac{1}{\omega} \cosh \omega u - \frac{1}{\omega} \sinh \omega u \right] \Big|_0^t$$

$$① \Rightarrow \int_0^t t \sin(hwu) du = \left[\frac{t \cosh wu}{w} \right]_0^t = \left(\frac{t \cosh wt}{w} - \frac{t}{w} \right)$$

$$② \Rightarrow \int_0^t u \sin(hwu) du = \left[(u) \cdot \frac{\cos(hwu)}{w} - (1) \frac{\sin(hwu)}{w^2} \right]_0^t$$

$$= t \cdot \frac{\cosh wt}{w} - \frac{\sinh wt}{w^2}$$

$$= \mathcal{L}^{-1} \left\{ \frac{w}{s^2(s^2 - w^2)} \right\} = \frac{t \cosh wt}{w} - \frac{t}{w} - \frac{t \cosh wt}{w} + \frac{\sinh wt}{w^2}$$

$$= \frac{\sinh wt}{w^2} - \frac{t}{w}$$

$$= \frac{\sinh wt}{w^2} - \frac{t}{w}$$

Q Find $\mathcal{L}^{-1} \left\{ \frac{40.5}{s^2} \right\}$

Q Find $\mathcal{L}^{-1} \left\{ \frac{5.5}{(s+1.5)(s-4)} \right\}$

Q Find $\mathcal{L}^{-1} \left\{ \frac{1}{(s^2+1)(s^2-1)} \right\}$

Ans: $\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$

$$F(s) = \frac{1}{s^2+1}$$

$$f(t) = \frac{\sin t}{1}$$

$$f(u) = \frac{\sin u}{1}$$

$$G(s) = \frac{1}{s^2-1}$$

$$g(t) = \frac{\sinh t}{1}$$

$$g(t-u) = \frac{\sinh(t-u)}{1}$$

$$= \int_0^t \frac{\sin \omega u}{\omega} \cdot \frac{\sinh \omega(t-u)}{\omega} du$$

$$= \int_0^t \sin u \cdot \sinh(t-u) du$$

$$= (\sin u) \cdot \left(\frac{\cosh(t-u)}{-1} \right) \Big|_0^t - \int_0^t \cos u \cdot \frac{\cosh(t-u)}{-1} du$$

$$= \frac{\sin t \cosh 0 - \sin 0 \cosh t}{-1} + \int_0^t \cos u \cosh(t-u) du = \frac{\sin t}{-1} + \int_0^t \cos u \cosh(t-u) du$$

$$= -\sin t + \int_0^t \cos u \cosh(t-u) du$$

$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

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$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

$$= \frac{t}{\omega} - \frac{t \cosh \omega t}{\omega}$$

Q.12

1. Find $\mathcal{L}^{-1} \left\{ \frac{5.5}{(s+1.5)(s-4)} \right\}$

Ans: $(5.5) \mathcal{L}^{-1} \left\{ \frac{1}{(s+1.5)(s-4)} \right\}$

$$\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

$$F(s) = \frac{1}{s+1.5} \quad f(t) = e^{-1.5t} \quad f(u) = e^{-1.5u}$$

$$G(s) = \frac{1}{s-4} \quad g(t) = e^{4t} \quad g(t-u) = e^{4(t-u)}$$

$$\mathcal{L}^{-1} \left\{ \frac{5.5}{(s+1.5)(s-4)} \right\} = 5.5 \int_0^t e^{-1.5u} \cdot e^{4(t-u)} du$$

$$= 5.5 \int_0^t e^{4t-4u-1.5u} du$$

$$= 5.5 \int_0^t e^{4t-5.5u} du$$

$$= 5.5 \cdot \left[\frac{e^{4t-5.5u}}{-5.5} \right]_0^t$$

$$= \frac{5.5}{-5.5} \cdot \left[e^{4t-5.5t} - e^{4t} \right]$$

$$= - \left[e^{-1.5t} - e^{4t} \right]$$

$$= \underline{\underline{e^{4t} - e^{-1.5t}}}$$

$$2. \quad \mathcal{L}^{-1} \left\{ \frac{2us}{(s^2 + \pi^2)^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{2us}{(s^2 + \pi^2)(s^2 + \pi^2)} \right\}$$

$$= 2 \mathcal{L}^{-1} \left\{ \frac{\bar{u}}{s^2 + \pi^2} \cdot \frac{s}{s^2 + \pi^2} \right\}$$

$$\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t f(u) g(t-u) du$$

$$F(s) = \sin \pi t \quad f(u) = \sin \pi u$$

$$G(s) = \cos \pi t \quad g(t-u) = \cos \pi (t-u)$$

$$\mathcal{L}^{-1} \left\{ \frac{2us}{(s^2 + \pi^2)^2} \right\} = 2 \int_0^t \sin(\pi u) \cdot \cos \pi (t-u) du$$

$$= 2 \int_0^t \frac{\sin(\pi u + \pi t - \pi u) + \sin(\pi u - \pi t + \pi u)}{2} du$$

$$= \int_0^t \sin(\pi t) + \sin(2\pi u - \pi t) du$$

$$= \left[-\frac{\cos(\pi t)}{\pi} + \frac{\cos(2\pi u - \pi t)}{2\pi} \right]_0^t$$

$$= \left[\sin(\pi t) \cdot u - \frac{\cos(2\pi u - \pi t)}{2\pi} \right]_0^t$$

$$= t \sin(\pi t) - \frac{\cos(2\pi t)}{2\pi} + \frac{\cos(\pi t)}{2\pi}$$

$$= \underline{\underline{t \sin(\pi t)}}$$

$$3. \quad \mathcal{L}^{-1} \left\{ \frac{18s}{(s^2+36)^2} \right\}$$

$$= 18 \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+36)} \cdot \frac{s}{(s^2+36)} \right\}$$

$$\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

$$F(s) = \frac{1}{s^2+36} \quad f(t) = \frac{\sin 6t}{6} \quad f(u) = \frac{\sin 6u}{6}$$

$$G(s) = \frac{s}{s^2+36} \quad g(t) = \cos 6t \quad g(t-u) = \cos(6t-6u)$$

$$\mathcal{L}^{-1} \left\{ \frac{18s}{(s^2+36)^2} \right\} = 18 \int_0^t \frac{\sin 6u}{6} \cdot \cos(6t-6u) du$$

$$= 3 \int_0^t \sin 6u \cdot \cos(6t-6u) du$$

$$= 3 \int_0^t \sin(6u+6t-6u) + \sin(6u-6t+6u) du$$

$$= 3 \left[u \sin 6t - \frac{\cos(12u-6t)}{12} \right]_0^t$$

$$= 3 \left[t \sin 6t - \frac{\cos 6t}{12} + \frac{\cos 6t}{12} \right]$$

$$= \underline{\underline{3t \sin 6t}}$$

$$Q. \quad \mathcal{L}^{-1} \left\{ \frac{240}{(s^2+1)(s^2+25)} \right\}$$

$$\text{Ans:} \quad 240 \mathcal{L}^{-1} \left\{ \frac{1}{s^2+1} \cdot \frac{1}{s^2+25} \right\}$$

$$\mathcal{L}^{-1} \{ F(s) \cdot G(s) \} = f(t) * g(t) = \int_0^t f(u) \cdot g(t-u) du$$

$$F(s) = \frac{1}{s^2+1} \quad f(t) = \sin t \quad f(u) = \sin u$$

$$G(s) = \frac{1}{s^2+25} \quad g(t) = \frac{\sin 5t}{5} \quad g(t-u) = \frac{\sin(5t-5u)}{5}$$

$$\mathcal{L}^{-1} \left\{ \frac{240}{(s^2+1)(s^2+25)} \right\} = 240 \int_0^t \sin u \cdot \frac{\sin(5t-5u)}{5} du$$

$$= 48 \int_0^t \sin u \cdot \sin(5t-5u) du$$

$$= 48 \int_0^t \frac{\cos(u-5t+5u) - \cos(u+5t-5u)}{2} du$$

$$= 24 \int_0^t \cos(6u-5t) - \cos(5t-4u) du$$

$$= 24 \left[\frac{\sin(6u-5t)}{6} - \frac{\sin(5t-4u)}{-4} \right]_0^t$$

$$= 24 \left[\frac{\sin t}{6} + \frac{\sin t}{4} + \frac{\sin 5t}{6} - \frac{\sin 5t}{4} \right]$$

$$= 24 \left[\frac{10 \sin t}{24} - \frac{2 \sin 5t}{24} \right]$$

$$= \frac{24}{24} (10 \sin t - 2 \sin 5t)$$

$$= \underline{10 \sin t - 2 \sin 5t}$$

$$5). \quad \mathcal{L}^{-1} \left\{ \frac{1}{(s-a)^2} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{(s-a)} \cdot \frac{1}{(s-a)} \right\}$$

$$F(s) = \frac{1}{s-a} \quad f(t) = e^{at} \quad f(u) = e^{au}$$

$$G(s) = \frac{1}{s-a} \quad g(t) = e^{at} \quad g(t-u) = e^{at-au}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s-a)^2} \right\} = \int_0^t e^{au} \cdot e^{at-au} du$$

$$= \left[\frac{e^{at}}{a} \right]_0^t$$

$$= \left[\frac{e^{at} - 1}{a} \right]$$

$$= e^{at} \int_0^t 1 du$$

$$= e^{at} [u]_0^t$$

$$= \underline{\underline{t \cdot e^{at}}}$$

$$6). \quad \mathcal{L}^{-1} \left\{ \frac{9}{s(s+3)} \right\}$$

$$F(s) = \frac{1}{s+3} \quad f(t) = e^{-3t} \quad f(u) = e^{-3u}$$

$$G(s) = \frac{1}{s} \quad g(t) = 1 \quad g(t-u) = 1$$

$$= 9 \int_0^t e^{-3u} \cdot 1 \, du$$

$$= 9 \left[\frac{e^{-3u}}{-3} \right]_0^t$$

$$= 9 \left(\frac{e^{-3t} - 1}{-3} \right)$$

$$= \frac{9(1 - e^{-3t})}{3} = \underline{\underline{3(1 - e^{-3t})}}$$

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$$\left(\frac{1 - e^{-3t}}{3} \right) =$$

$$\text{sub } t \rightarrow 0 \Rightarrow$$

$$3(1 - e^{-3 \cdot 0}) =$$

$$3(1 - 1) =$$

Laplace Transform of Derivatives

$$\text{If } L\{f(t)\} = F(s)$$

$$\text{then } L\{f'(t)\} = s \cdot F(s) - f(0)$$

$$L\{f''(t)\} = s^2 \cdot F(s) - s f(0) - f'(0)$$

$$L\{f'''(t)\} = s^3 \cdot F(s) - s^2 f(0) - s f'(0) - f''(0)$$

⋮

$$\text{In general, } L\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

Q solve $y'' + y' - 6y = 0$ given that $y(0) = 1$, $y'(0) = 1$

Note: [If $y(0)$ and $y'(0)$ is not given then don't consider the as is wrong ~~take~~ assume any value and do]

[This question can be asked in two way,

$$(i) y'' + y' - 6y = 0 \text{ OR } (ii) \frac{d^2y}{dt^2} + \frac{dy}{dt} - 6y = 0]$$

$$\text{Ans: } y'' + y' - 6y = 0$$

$$L\{y''\} + L\{y'\} - 6L\{y\} = L\{0\}$$

$$L\{f''(t)\} = s^2 F(s) - s f(0) - f'(0)$$

$$\therefore L\{y''(t)\} = s^2 Y(s) - s y(0) - y'(0)$$

$$L\{y'(t)\} = s Y(s) - y(0)$$

$$L\{y(t)\} = Y(s)$$

$$2 \quad \cancel{s^2 F(s)} = 0$$

$$2 \quad s^2 Y(s) - s y(0) - y'(0) + s Y(s) - y(0) - 6 Y(s) = 0$$

$$2 \quad Y(s) [s^2 + s - 6] - s - 1 - 1 = 0$$

$$Y(s) [s^2 + s - 6] - s - 2 = 0$$

$$Y(s) = \frac{s+2}{(s^2+s-6)}$$

$$\mathcal{L}^{-1} \left\{ \frac{s+2}{s^2+s-6} \right\} = \mathcal{L}^{-1} \left\{ \frac{s+2}{s^2+s-6 + \frac{1-1}{2} + \frac{1-1}{2}} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+3)(s-2)} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+3)(s-2)} \right\}$$

$$= A \mathcal{L}^{-1} \left\{ \frac{1}{s+3} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s-2} \right\}$$

$$\therefore \frac{s+2}{(s+3)(s-2)} = \frac{A}{(s+3)} + \frac{B}{(s-2)}$$

$$\therefore s+2 = A(s-2) + B(s+3)$$

$$\text{put } s = -2$$

$$B = \frac{4}{5}$$

$$\text{put } s = 2$$

$$A = \frac{1}{5}$$

$$\therefore \mathcal{L}^{-1} \left\{ \frac{s+2}{(s+3)(s-2)} \right\} = \frac{1}{5} e^{-3t} + \frac{4}{5} e^{2t}$$

$$y(t) = \frac{1}{5} (e^{-3t} + 4e^{2t})$$

Q. solve $y'' - y = t$ where $y(0) = 1, y'(0) = 1$

Ans. $\mathcal{L}\{y'' - y\} = \mathcal{L}\{t\}$

$$\mathcal{L}\{y''\} - \mathcal{L}\{y\} = \mathcal{L}\{t\}$$

$$s^2 Y(s) - s y(0) - y'(0) - Y(s) = \frac{1}{s^2} \therefore \mathcal{L}\{y''(t)\} = s^2 Y(s) - s y(0) - y'(0)$$

$$s^2 Y(s) - s - 1 - Y(s) = \frac{1}{s^2} \therefore \mathcal{L}\{t\} = \frac{1}{s^2}$$

$$(s^2 - 1) Y(s) = \frac{1}{s^2} + s + 1$$

$$Y(s) = \frac{1}{s^2 - 1} + \frac{s+1}{s^2 - 1}$$

$$\frac{(1+s^3+s^2)}{(s^4-s^2)}$$

Solution $y(t) = \mathcal{L}^{-1}[Y(s)]$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s^2(s^2-1)} \right\} + \mathcal{L}^{-1} \left\{ \frac{s}{s^2-1} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{s^2-1} \right\}$$

$$= \mathcal{L}^{-1} \left\{ \frac{1}{s^2(s^2-1)} \right\} + \cosh t + \sinh t$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2(s-1)} \right\} = \int_0^t \int_0^t f(\tau) d\tau dt$$

$$= \int_0^t \int_0^t \sinh \tau d\tau dt$$

$$= \int_0^t [\cosh \tau]_0^t dt$$

$$= \int_0^t (\cosh t - \cosh 0) dt$$

$$= \int_0^t \cosh t - 1 dt$$

$$= [\sinh t - t]_0^t$$

$$= (\sinh t - t - \sinh 0 + 0)$$

$$= \sinh t - t$$

$$\therefore \text{solution } y(t) = \sinh t - t + \cosh t + \sinh t$$

$$= \underline{\underline{2\sinh t + \cosh t - t}}$$

$$Q. \text{ solve } y' + \frac{2}{3}y = -4 \cos 2t$$

$$\frac{1+2+\frac{1}{3}}{1-\frac{1}{3}} = (2) \quad y(0) = 20$$

$$\text{Ans: } \mathcal{L}\{y' + \frac{2}{3}y\} = \mathcal{L}\{-4 \cos 2t\}$$

$$\mathcal{L}\{y'\} + \frac{2}{3}\mathcal{L}\{y\} = \mathcal{L}\{-4 \cos 2t\}$$

$$sY(0) - y(0) + \frac{2}{3}Y(0) = -4 \left[\frac{s}{s^2 + 4} \right]$$

$$Y(0) \left(s + \frac{2}{3} \right) = \frac{-4s}{s^2 + 4}$$

$$Y(s) = \frac{-4s}{s^2+4} \times \frac{1}{(s+\frac{2}{3})}$$

$$= \frac{-4s}{(s^2+4)(s+\frac{2}{3})}$$

$$\text{solution } y(t) = \mathcal{L}^{-1}\{Y(s)\} = -4 \mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)(s+\frac{2}{3})}\right\}$$

apply convolution theorem;

$$F(s) = \frac{s}{s^2+4} \quad f(t) = \frac{\cos 2t}{2} \quad f(u) = \frac{\cos 2u}{2}$$

$$G(s) = \frac{1}{s+\frac{2}{3}} \quad g(t) = e^{-\frac{2}{3}t} \quad g(t-u) = e^{-\frac{2}{3}(t-u)}$$

$$\mathcal{L}^{-1}\{F(s) \cdot G(s)\} = f * g = \int_0^t f(u) \cdot g(t-u) du$$

$$= \int_0^t \cos 2u \cdot e^{-\frac{2}{3}(t-u)} du$$

$$= \int_0^t \cos 2u \cdot e^{-\frac{2}{3}t} \cdot e^{\frac{2}{3}u} du$$

$$= e^{-\frac{2}{3}t} \int_0^t \cos 2u \cdot e^{\frac{2}{3}u} du$$

$$= e^{-\frac{2}{3}t} \left[\frac{e^{\frac{2}{3}u}}{\left(\frac{4}{9}+4\right)} \left[\frac{2}{3} \cos 2u + 2 \sin 2u \right] \right]_0^t$$

$$= e^{-\frac{2}{3}t} \left[\frac{e^{\frac{2}{3}t}}{\left(\frac{4}{9}+4\right)} \left[\frac{2}{3} \cos 2t + 2 \sin 2t \right] - \frac{e^0}{\left(\frac{4}{9}+4\right)} \left[\frac{2}{3} \cos 0 + 0 \right] \right]$$

$$= \frac{9e^{-\frac{2}{3}t}}{40} \left[e^{\frac{2}{3}t} \left[\frac{2}{3} \cos 2t + 2 \sin 2t \right] - \frac{2}{3} \right]$$

$$= \frac{9}{40} e^{-2/3 t} \left[e^{2/3 t} \left[\frac{2}{3} \cos 2t + 2 \sin 2t \right] - \frac{2}{3} \right]$$

$$\therefore \text{solution } y(t) = -\frac{9}{40} \times \frac{9}{40} e^{-2/3 t} \left[e^{2/3 t} \left[\frac{2}{3} \cos 2t + 2 \sin 2t \right] - \frac{2}{3} \right]$$

$$= -\frac{9}{10} e^{-2/3 t} \left[e^{2/3 t} \left[\frac{2}{3} \cos 2t + 2 \sin 2t \right] - \frac{2}{3} \right]$$

Q. solve $y'' - \frac{1}{4}y = 0$, $y(0) = 12$, $y'(0) = 0$

Ans. $\mathcal{L}\{y'' - \frac{1}{4}y\} = \mathcal{L}\{0\}$

$$\mathcal{L}\{y''\} = -\frac{1}{4}\mathcal{L}\{y\} = 0$$

$$s^2 Y(s) - s y(0) - y'(0) - \frac{1}{4} Y(s) = 0$$

$$Y(s) (s^2 - \frac{1}{4}) - 12s - 0 = 0$$

$$Y(s) = \frac{12s}{(s^2 - \frac{1}{4})}$$

$$\mathcal{L}^{-1}\{Y(s)\} = 12 \mathcal{L}^{-1}\left\{\frac{s}{s^2 - \frac{1}{4}}\right\}$$

$$\underline{\underline{y(t) = 12 \cdot \cosh \frac{1}{2} t}}$$

$\therefore$ ~~$y(t) = 12 \cdot \cosh \frac{1}{2} t$~~

Q. solve: $y'' + y' + 9y = 0$, $y(0) = 0.16$, $y'(0) = 0$

an. $\mathcal{L}\{y'' + y' + 9y\} = \mathcal{L}\{0\}$

$$\mathcal{L}\{y''\} + \mathcal{L}\{y'\} + 9\mathcal{L}\{y\} = 0$$

$$s^2 Y(s) - sy(0) - y'(0) + sY(s) - y(0) + 9Y(s) = 0$$

$$Y(s) [s^2 + s + 9] - sy(0) - y'(0) - y(0) = 0$$

$$Y(s) [s^2 + s + 9] = 0.16s + 0.16$$

$$Y(s) = \frac{0.16s + 0.16}{s^2 + s + 9}$$

$$y(s) = 0.16 \mathcal{L}^{-1} \left\{ \frac{s+1}{s^2+s+9} \right\} \quad s^2+s+9 = s^2+s+9+\frac{1-1}{4}$$

$$= 0.16 \mathcal{L}^{-1} \left\{ \frac{s+\frac{1}{2}-\frac{1}{2}+1}{\left(s+\frac{1}{2}\right)^2 + \frac{35}{4}} \right\}$$

$$= 0.16 \left\{ \frac{\left(s+\frac{1}{2}\right) + \frac{1}{2}}{\left(s+\frac{1}{2}\right)^2 + \frac{35}{4}} \right\}$$

$$= 0.16 \left[e^{-1/2t} \cos \frac{\sqrt{35}}{2} t + \frac{1}{2} \frac{\sin \frac{\sqrt{35}}{2} t}{\frac{\sqrt{35}}{2}} \right]$$

$$= \left[0.16 e^{-1/2t} \left[\cos \frac{\sqrt{35}}{2} t + \frac{\sin \frac{\sqrt{35}}{2} t}{\frac{\sqrt{35}}{2}} \right] \right]$$

$$\left[\frac{1}{p+2} \right] \frac{1}{(p+2)^2 + \frac{35}{4}} \left[\frac{1}{(p+2)} \right]$$

$$Q. y'' + 7y' + 12y = 21e^{3t} \quad y(0) = 3.5, \quad y'(0) = -10$$

$$Ans: \mathcal{L}\{y''\} + 7\mathcal{L}\{y'\} + 12\mathcal{L}\{y\} = 21\mathcal{L}\{e^{3t}\}$$

$$s^2 Y(s) - s y(0) - y'(0) + 7(s Y(s) - y(0)) + 12 Y(s) = \frac{21}{s-3}$$

$$s^2 Y(s) - s y(0) - y'(0) + 7s Y(s) - 7 y(0) + 12 Y(s) = \frac{21}{s-3}$$

$$Y(s) [s^2 + 7s + 12] - 3.5s + 10 - 7(3.5) = \frac{21}{s-3}$$

$$Y(s) = \frac{\frac{21}{s-3} + 3.5s - 10 + 14.5}{s^2 + 7s + 12}$$

$$\therefore Y(s) = \frac{21}{(s-3)(s^2+7s+12)} + \frac{3.5s + 14.5}{(s^2+7s+12)}$$

$$= \frac{21}{(s-3)(s+3)(s+4)} + \frac{3.5s + 14.5}{(s+3)(s+4)}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{21}{(s-3)(s+3)(s+4)} \right\} + \mathcal{L}^{-1} \left\{ \frac{3.5s + 14.5}{(s+3)(s+4)} \right\}$$

$$= 21 \left[A \cdot \mathcal{L}^{-1} \left\{ \frac{1}{(s-3)} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s+3} \right\} + C \mathcal{L}^{-1} \left\{ \frac{1}{(s+4)} \right\} \right] +$$

$$\left[A \mathcal{L}^{-1} \left\{ \frac{1}{(s+3)} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s+4} \right\} \right]$$

$$\frac{1}{(s-3)(s+3)(s+4)} = \frac{A}{(s-3)} + \frac{B}{(s+3)} + \frac{C}{(s+4)}$$

$$1 = A(s+3)(s+4) + B(s-3)(s+4) + C(s-3)(s+3)$$

put $s = -3$ put $s = 3$

$$1 = B(-6)(1)$$

$$1 = 42A$$

$$B = \frac{-1}{6}$$

$$A = \frac{1}{42}$$

put $s = -4$

$$1 = C(7)(-1)$$

$$C = \frac{1}{7}$$

$$\frac{2-5s+14-5}{(s+3)(s+4)} = \frac{A}{(s+3)} + \frac{B}{(s+4)}$$

$$\therefore A = 4 \quad B = \frac{-1}{2}$$

$$\therefore y(t) = 21 \left[\frac{1}{42} e^{3t} - \frac{1}{6} e^{-3t} + \frac{1}{7} e^{-4t} \right] + 4e^{-3t} - \frac{1}{2} e^{-4t}$$

$$= \frac{1}{2} e^{3t} - \frac{21}{6} e^{-3t} + 3e^{-4t} + 4e^{-3t} - \frac{1}{2} e^{-4t}$$

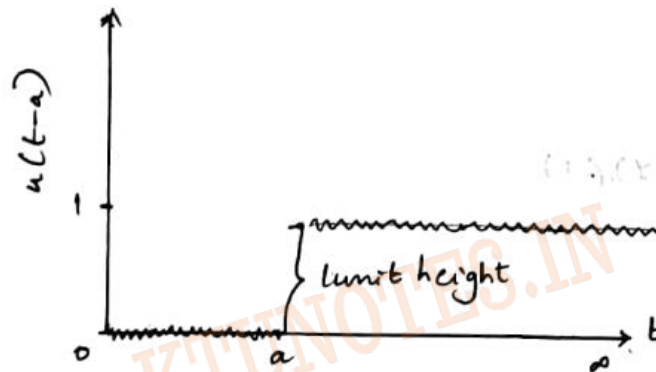
$$= \frac{1}{2} e^{3t} + \frac{5}{2} e^{-4t} + \frac{1}{2} e^{-3t}$$

Unit Step Function (Heavisides Function)

(study proof)

The unit step function or Heaviside function is denoted by $u(t-a)$ where a is a positive number, defined as ~~$u(t-a)$~~

$$u(t-a) = \begin{cases} 0 & t < a \\ 1 & t > a \end{cases}$$



Laplace Transform of $u(t-a)$

$$L\{u(t-a)\} = \int_0^{\infty} e^{-st} \cdot u(t-a) dt$$

$$= \int_0^a e^{-st} u(t-a) dt + \int_a^{\infty} e^{-st} \cdot u(t-a) dt$$

$$= \int_0^a e^{-st} \cdot 0 dt + \int_a^{\infty} e^{-st} \cdot 1 dt$$

$$= \int_a^{\infty} e^{-st} dt$$

$$= \int \frac{[e^{-st}]_a^\infty}{-s} a$$

$$= \left[\frac{e^{-sa}}{-s} - \frac{e^{-as}}{-s} \right] = \frac{e^{-as}}{s}$$

$$\therefore \boxed{L\{u(t-a)\} = \frac{e^{-as}}{s}}$$

Second shifting Theorem (t-shifting) multiplication by unit step function

$$\checkmark L\{f(t-a) \cdot u(t-a)\} = e^{-as} \cdot F(s)$$

$$\checkmark L^{-1}\{e^{-as} \cdot F(s)\} = f(t-a) u(t-a)$$

Q Find $L\{(t-3) \cdot u(t-3)\}$

Ans: $L\{(t-3) u(t-3)\} = e^{-3s} \cdot F(s)$

$$f(t-3) = t-3$$

$$f(t) = t \quad \text{for } t \geq 3$$

$$\therefore F(s) = L\{f(t)\} = L\{t\} = \frac{1}{s^2}$$

$$\therefore L\{(t-3) u(t-3)\} = \frac{e^{-3s}}{s^2}$$

$$L\{(t-3) u(t-3)\} = \frac{e^{-3s}}{s^2}$$

Q. $\mathcal{L}\{\sin t \cdot u(t-\pi)\}$

Ans: $\mathcal{L}\{-\sin(t-\pi) \cdot u(t-\pi)\}$

$\therefore = -\mathcal{L}\{\sin(t-\pi) \cdot u(t-\pi)\}$

$= -e^{-\pi s} F(s)$

$= -e^{-\pi s} \cdot \mathcal{L}\{\sin t\}$

$= -e^{-\pi s} \cdot \frac{1}{s^2+1}$

$= \frac{-e^{-\pi s}}{s^2+1}$

~~$f(t-\pi) = \sin(t-\pi)$~~
 $f(t) = \sin t$

(Imp)

Q. $\mathcal{L}\{t^4 \cdot u(t)\}$

Ans: $\mathcal{L}\{t^4 \cdot u(t-2)\}$

$\mathcal{L}\{(t-2+2)^4 \cdot u(t-2)\}$

$\mathcal{L}\{[(t-2)+2]^4 \cdot u(t-2)\}$

$\mathcal{L}\{[(t-2)^2 + 4 + 4(t-2)][(t-2)^2 + 4 + 4(t-2)] \cdot u(t-2)\}$

$= \mathcal{L}\{[(t-2)^4 + 4(t-2)^2 + 4(t-2)^3 + 4(t-2)^2 + 16 + 16(t-2) + 4(t-2)^3 + 16(t-2) + 16(t-2)^2] u(t-2)\}$

$$= \mathcal{L}\{((t-2)^4 + 4(t-2)^3 \cdot 2 + 6(t-2)^2 \cdot 2^2 + 4(t-2) \cdot 2^3 + 2^4) u(t-2)\}$$

$$= \mathcal{L}\{((t-2)^4 + 8(t-2)^3 + 24(t-2)^2 + 32(t-2) + 16) u(t-2)\}$$

$$= \mathcal{L}\{(t-2)^4 u(t-2)\} + \mathcal{L}\{8(t-2)^3 u(t-2)\} + \mathcal{L}\{24(t-2)^2 u(t-2)\} + 32 \mathcal{L}\{(t-2) u(t-2)\} + \mathcal{L}\{16 u(t-2)\}$$

$$= e^{-2s} \cdot \frac{4!}{s^5} + (8e^{-2s}) \frac{3!}{s^4} + 24e^{-2s} \frac{2!}{s^3} + \frac{32e^{-2s}}{s^2} + \frac{16e^{-2s}}{s}$$

$$= e^{-2s} \left[\frac{24}{s^5} + \frac{8 \cdot 3!}{s^4} + \frac{48}{s^3} + \frac{32}{s^2} + \frac{16}{s} \right]$$

$$\underline{\underline{(s-3)^3 (s-3)^4 = \left\{ \frac{(s-3)^7}{s^2} \right\}}}$$

$$(s-3)^3 \frac{(s-3)^4}{10} =$$

$$\left\{ \frac{(2s-5-1)^4}{s^2} \right\} \text{ is not}$$

$$\left\{ \frac{(2s-5-1)^4}{s^2} \right\} \text{ is not}$$

$$\left[\left\{ \frac{(2s-5)}{s^2} \right\} \right] \text{ is } \left[\frac{1}{s^2} \right] \text{ is not}$$

Q. Find $\mathcal{L}^{-1}$ of $\left\{ \frac{e^{-2s}}{s} \right\}$

[If numerator is an exponential function apply 2nd shifting theorem]

Ans. $\mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s} \right\} = f(t-a) u(t-a)$
 $= \underline{u(t-2)}$

$\mathcal{L}^{-1} \left\{ \frac{e^{-as}}{s} \right\} = u(t-a)$

Q. Find $\mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s^6} \right\}$

$\mathcal{L}^{-1} \left\{ e^{-as} \cdot F(s) \right\} = f(t-a) u(t-a)$

$F(s) = \frac{1}{s^6}$

$f(t) = \frac{t^5}{5!}$ $\therefore f(t-a) = \frac{(t-2)^5}{5!}$

$\mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s^6} \right\} = f(t-2) u(t-2)$
 $= \frac{(t-2)^5}{5!} u(t-2)$

Q. Find $\mathcal{L}^{-1} \left\{ \frac{4(1-e^{-7s})}{s^2+4} \right\}$

Ans. $4 \mathcal{L}^{-1} \left\{ \frac{1-e^{-7s}}{s^2+4} \right\}$

$= 4 \left[\mathcal{L}^{-1} \left\{ \frac{1}{s^2+4} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-7s}}{s^2+4} \right\} \right]$

$$\mathcal{L}^{-1}\{e^{-as} F(s)\} = f(t-a) u(t-a)$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+4}\right\} = f(t-\pi) u(t-\pi)$$

$$F(s) = \frac{1}{s^2+4}$$

$$f(t) = \frac{\sin 2t}{2}$$

$$f(t-\pi) = \frac{\sin(2t-2\pi)}{2} = \frac{-\sin 2t}{2}$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-\pi s}}{s^2+4}\right\} = \frac{-\sin 2t}{2} u(t-\pi)$$

$$\mathcal{L}^{-1}\left\{\frac{4(1-e^{-\pi s})}{s^2+4}\right\} = \frac{4\sin 2t}{2} + \frac{2\sin 2t}{2} u(t-\pi)$$

$$= 2\sin 2t + 2\sin 2t u(t-\pi)$$

Find $\mathcal{L}^{-1}\left\{\frac{(1+e^{-2\pi(s+1)})}{(s+1)^2+1}\right\}$

$$= \mathcal{L}^{-1}\left\{\frac{(s+1)}{(s+1)^2+1} + \frac{(s+1)e^{-2\pi(s+1)}}{(s+1)^2+1}\right\}$$

$$= e^{-t} \cos t + e^{-t} \mathcal{L}^{-1}\left\{\frac{e^{-2\pi s}}{s^2+1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{e^{-2\pi s}}{s^2+1}\right\} = f(t-2\pi) u(t-2\pi)$$

$$F(s) = \frac{1}{s^2+1} \quad f(t) = \sin t \quad f(t-2\pi) = \sin(t-2\pi)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+1} (1 + e^{-2\pi s})\right\}$$

$$\mathcal{L}^{-1}\left\{(1 + e^{-2\pi s}) \left(\frac{s+1}{(s+1)^2+1}\right)\right\}$$

$$= e^{-t} \cos t + e^{-t} \cos(t-2\pi) u(t-2\pi)$$

$$= e^{-t} \left[\cos t + \cos(t-2\pi) u(t-2\pi) \right]$$

Q. Find the inverse transform of $F(s) = \frac{e^{-s}}{(s^2+\pi^2)} + \frac{e^{-2s}}{(s^2+\pi^2)} + \frac{e^{-3s}}{(s+2)^2}$

Ans: $= \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2}\right\} + \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{(s^2+\pi^2)}\right\} + \mathcal{L}^{-1}\left\{\frac{e^{-3s}}{(s+2)^2}\right\}$

① $\mathcal{L}^{-1}\left\{\frac{1}{s^2+\pi^2}\right\} = \frac{1}{\pi} \sin \pi t$

① $\Rightarrow$

$$\mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2}\right\} = \frac{1}{\pi} \sin \pi(t-1) u(t-1)$$

$$F(s) = \frac{1}{s^2+\pi^2} \quad f(t) = \frac{\sin \pi t}{\pi} \quad f(t-1) = \frac{\sin \pi(t-1)}{\pi}$$

$$\therefore \mathcal{L}^{-1}\left\{\frac{e^{-s}}{s^2+\pi^2}\right\} = \frac{\sin \pi(t-1)}{\pi} u(t-1)$$

Q 27.

$$\mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s^2 + \pi^2} \right\} = f(t-2) u(t-2)$$

$$F(s) = \frac{1}{s^2 + \pi^2} \quad f(t) = \frac{\sin \pi t}{\pi} \quad f(t-2) = \frac{\sin \pi(t-2)}{\pi}$$

$$\mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s^2 + \pi^2} \right\} = \frac{\sin \pi(t-2)}{\pi} u(t-2)$$

Q 28.

$$\mathcal{L}^{-1} \left\{ \frac{e^{-3s}}{(s+2)^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{e^{-3s}}{(s+2)^2} \right\}$$

$$= e^{-2t} f(t-3) u(t-3)$$

$$F(s) = \frac{1}{(s+2)^2} \quad f(t) = t \cdot e^{-2t} \quad f(t-3) = (t-3) \cdot e^{-2(t-3)}$$

$$\mathcal{L}^{-1} \left\{ \frac{e^{-3s}}{(s+2)^2} \right\} = e^{-2(t-3)} (t-3) u(t-3)$$

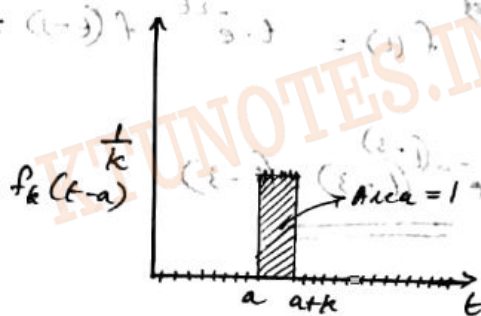
$$\mathcal{L}^{-1} \{ f(s) \} = \frac{\sin \pi(t-1)}{\pi} u(t-1) + \frac{\sin \pi(t-2)}{\pi} (t-2) u(t-2) + e^{-2(t-3)} (t-3) u(t-3)$$

Unit Impulse Function (study proof) (Dirac Delta Function)

In machines we come across problems where a very large force acts for a very small time or large pressure acting over a small area. To deal with such problems we introduce the unit impulse function.

Define a function,

$$\text{Force Function} \rightarrow f_k(t-a) = \begin{cases} \frac{1}{k} & a \leq t \leq a+k \\ 0 & \text{otherwise} \end{cases}$$



Note: 'a' is a +ve number and 'k' an arbitrary small +ve real number.

Integrating this function we get,

$$\begin{aligned} & \int_0^{\infty} f_k(t-a) dt \\ &= \int_0^{a+k} \frac{1}{k} dt \\ &= \frac{1}{k} (t)_0^{a+k} = \frac{1}{k} (a+k-0) = 1 \end{aligned}$$

Definition of Unit Impulse Function.

Let $\delta(t-a)$ be a function of t defined for all the values of t and a is any +ve number, then the unit impulse function about a is denoted by,

$$\delta(t-a) = \begin{cases} \infty, & t=a \\ 0, & \text{otherwise} \end{cases}$$

provided $\int_0^{\infty} \delta(t-a) dt = 1$

note 1: $\delta(t-a) = \lim_{k \rightarrow 0} f_k(t-a)$

note 2: Mean value Theorem (MVT)

If $f(x)$ is a function continuous at $x=a$ and $a < c < b$ then,

$$\int_a^b f(x) dx = (b-a) f(c)$$

Laplace Transform of Unit Impulse Function.

$$\boxed{L\{\delta(t-a)\} = e^{-as}}$$

proof:

$$\int_0^{\infty} f(t) \cdot \delta(t-a) dt = \int_a^{a+k} f(t) \cdot \frac{1}{k} dt$$

$$= \frac{1}{k} \int_a^{a+k} f(t) dt$$

$f(t)$ is continuous at all pts.

$$\lim_{k \rightarrow 0} \frac{1}{k} \int_a^{a+k} f(t) dt \rightarrow \text{Apply MVT}$$

$$f(t) = f(0) \quad \text{--- (1)}$$

Let $t \rightarrow 0$, $c = a$ ($a < c < a$) & $f(t-a) \rightarrow f(t-a)$ for
 $\int_0^\infty f(t) \delta(t-a) dt = f(a) \quad \text{--- (2)}$

choose $f(t) = e^{-st}$ $\therefore f(t) = e^{-st}$
 $f(a) = e^{-as}$
 $\int_0^\infty e^{-st} \delta(t-a) dt = e^{-as}$

$$\therefore \mathcal{L}\{\delta(t-a)\} = e^{-as}$$

Solution of D.Es involving Unit step & Dirac Delta

Q. solve $y'' + 3y' + 2y = u(t-1) - u(t-2)$, $y(0) = 0$, $y'(0) = 0$

Ans: $\mathcal{L}\{y''\} + 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{u(t-1)\} - \mathcal{L}\{u(t-2)\}$

$$s^2 Y(s) - s y(0) - y'(0) + 3s Y(s) - 3y(0) + 2Y(s) = \frac{e^{-s}}{s} - \frac{e^{-2s}}{s}$$

$$Y(s) (s^2 + 3s + 2) = \frac{e^{-s} - e^{-2s}}{s}$$

$$Y(s) = \frac{e^{-s} - e^{-2s}}{s(s^2 + 3s + 2)}$$

$$= \frac{e^{-s}}{s(s+1)(s+2)} - \frac{e^{-2s}}{s(s+1)(s+2)}$$

$$= \frac{e^{-s}}{s(s+1)(s+2)} - \frac{e^{-2s}}{s(s+1)(s+2)}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{e^{-s}}{s(s+1)(s+2)} \right\} - \mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s(s+1)(s+2)} \right\}$$

$$= \mathcal{L}^{-1} \left\{ e^{-s} \cdot \frac{1}{s(s+1)(s+2)} \right\} - \mathcal{L}^{-1} \left\{ e^{-2s} \cdot \frac{1}{s(s+1)(s+2)} \right\}$$

$$= f(t-1) u(t-1) - f(t-2) u(t-2)$$

for (1) and (2) $\Rightarrow$

$$F(s) = \frac{1}{s(s+1)(s+2)}$$

$$f(t) = A \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\} + C \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\}$$

$$\frac{1}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2}$$

$$1 = A(s+1)(s+2) + B(s)(s+2) + C(s)(s+1)$$

put $s=0$

$$A = \frac{1}{2}$$

put $s=-2$

$$C = \frac{1}{2}$$

put $s=-1$

$$B = -1$$

$$\therefore f(t) = \frac{1}{2} - e^{-t} + \frac{1}{2} e^{-2t}$$

$$f(t-1) = \frac{1}{2} - e^{-(t-1)} + \frac{1}{2} e^{-2(t-1)}$$

$$f(t-2) = \frac{1}{2} - e^{-(t-2)} + \frac{1}{2} e^{-2(t-2)}$$

$$y(t) = \left[\frac{1}{2} - e^{-(t-1)} + \frac{1}{2} e^{-2(t-1)} \right] u(t-1) - \left[\frac{1}{2} - e^{-(t-2)} + \frac{1}{2} e^{-2(t-2)} \right] u(t-2)$$

Q. $y'' + 3y' + 2y = f(t-1)$, $y(0) = 0$, $y'(0) = 0$

Ans $\mathcal{L}\{y''\} + 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{f(t-1)\}$

$$s^2 Y(s) - s y(0) - y'(0) + 3s Y(s) - y(0) + 2Y(s) = e^{-s}$$

$$Y(s) [s^2 + 3s + 2] = e^{-s}$$

$$Y(s) = \frac{e^{-s}}{s^2 + 3s + 2}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{e^{-s}}{(s+2)(s+1)} \right\}$$

$$= \mathcal{L}^{-1} \left\{ e^{-s} \cdot \frac{1}{(s+2)(s+1)} \right\}$$

$$= f(t-1) u(t-1)$$

$$F(s) = \frac{1}{(s+1)(s+2)}$$

$$f(t) = A \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} + B \mathcal{L}^{-1} \left\{ \frac{1}{s+1} \right\}$$

$$\therefore 1 = A(s+1) + B(s+2)$$

put $s = -1$

$$B = 1$$

put $s = -2$

$$A = -1$$

$$\therefore f(t) = e^{-t} - e^{-2t}$$

$$f(t-1) = \frac{1}{e} (e^{-(t-1)} - e^{-2(t-1)})$$

$$\therefore y(t) = \underline{\underline{[e^{-(t-1)} - e^{-2(t-1)}] u(t-1)}}$$

Laplace Transforms

$$\mathcal{L}\{e^{at} F(s)\} = F(s-a)$$

$$\mathcal{L}\{t \cdot f(t)\} = -\frac{d}{ds} F(s)$$

$$\mathcal{L}\left\{\frac{1}{t} \cdot f(t)\right\} = \int_s^\infty F(s) ds$$

$$\mathcal{L}\left\{\int_0^t f(t) dt\right\} = \frac{1}{s} \cdot F(s)$$

$$\mathcal{L}\{f''(s)\} = s^2 F(s) - s f(0) - f'(0)$$

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$$

$$\mathcal{L}\{f(t-a) u(t-a)\} = e^{-as} \cdot F(s)$$

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

Inverse

$$\mathcal{L}^{-1}\{F(s-a)\} = e^{at} f(t)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s} \cdot F(s)\right\} = \int_0^t f(t) dt$$

$$\mathcal{L}^{-1}\{e^{-as} F(s)\} = f(t-a) u(t-a)$$

Convolution Theorem

$$\mathcal{L}\{F(s) \cdot G(s)\} = f(t) * g(t) = \int_0^t f(\tau) \cdot g(t-\tau) d\tau$$

Q. solve $y'' + 2y' + 2y = [1 - u(t-2)]e^t - e^2 \delta(t-2)$

$y(0) = 0, \quad y'(0) = 1$

Ans: $\mathcal{L}\{y''\} + 2\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{e^t\} - \mathcal{L}\{e^t \cdot u(t-2)\} - \mathcal{L}\{e^2 \cdot \delta(t-2)\}$

LHS $\Rightarrow$

$$\begin{aligned} \Rightarrow s^2 Y(s) - s y(0) - y'(0) + 2s Y(s) - 2y(0) + 2Y(s) \\ = Y(s)(s^2 + 2s + 2) - 1 \end{aligned}$$

RHS $\Rightarrow$

$$\begin{aligned} \mathcal{L}\{e^t\} - \mathcal{L}\{e^t \cdot u(t-2)\} - \mathcal{L}\{e^2 \cdot \delta(t-2)\} \\ = \frac{1}{(s-1)} - \frac{e^{-2(s-1)}}{(s-1)} - e^2 \cdot e^{-2s} \end{aligned}$$

equating LHS = RHS

$$Y(s)(s^2 + 2s + 2) - 1 = \frac{1}{(s-1)} - \frac{e^{-2(s-1)}}{(s-1)} - e^2 \cdot e^{-2s}$$

$$Y(s) = \frac{1}{(s-1)(s^2+2s+2)} - \frac{e^{-2(s-1)}}{(s-1)(s^2+2s+2)} - \frac{e^2 \cdot e^{-2s}}{(s^2+2s+2)} + \frac{1}{(s^2+2s+2)}$$

$$y(t) = \mathcal{L}^{-1}\left\{ \frac{1}{(s-1)(s^2+2s+2)} \right\} - \mathcal{L}^{-1}\left\{ \frac{e^{-2(s-1)}}{(s-1)(s^2+2s+2)} \right\} - \mathcal{L}^{-1}\left\{ \frac{e^2 \cdot e^{-2s}}{(s^2+2s+2)} \right\} + \mathcal{L}^{-1}\left\{ \frac{1}{(s^2+2s+2)} \right\}$$

$$s^2 + 2s + 2 = (s+1)^2 + 1$$

$$= \frac{1}{(s+1)^2 + 1} = \frac{1}{(s+1)^2 + 1^2} = \frac{1}{(s+1)^2 + 1^2}$$

solving ①

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s-1)(s^2+2s+2)} \right\} = A \mathcal{L}^{-1} \left\{ \frac{1}{(s-1)} \right\} + \mathcal{L}^{-1} \left\{ \frac{Bs+C}{s^2+2s+2} \right\}$$

$$1 = A(s^2+2s+2) + (Bs+C)(s-1)$$

put $s = 1$

$$A = \frac{1}{5}$$

put $s = 0$

$$C = -\frac{3}{5}$$

put $s = -1$

$$B = -\frac{1}{5}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s-1)(s^2+2s+2)} \right\} = \frac{1}{5} e^t + \mathcal{L}^{-1} \left\{ \frac{-\frac{1}{5}s - \frac{3}{5}}{s^2+2s+2} \right\}$$

$$= \frac{e^t}{5} - \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{s+3}{s^2+2s+2} \right\}$$

$$= \frac{e^t}{5} - \frac{1}{5} \mathcal{L}^{-1} \left\{ \frac{(s+1) + 2}{(s+1)^2 + 1} \right\}$$

$$= \frac{e^t}{5} - \frac{e^{-t}}{5} \cos t - \frac{2e^{-t}}{5} \sin t$$

solving ②

$$\mathcal{L}^{-1} \left\{ \frac{e^{-2(s-1)}}{(s-1)(s^2+2s+2)} \right\} = \mathcal{L}^{-1} \left\{ \frac{e^2 \cdot e^{-2s}}{(s-1)(s^2+2s+2)} \right\}$$

$$= e^2 \mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{(s-1)(s^2+2s+2)} \right\}$$

$$= e^2 [f(t-2) u(t-2)]$$

$$= e^2 \left[\frac{e^{t-2}}{5} - \frac{e^{-(t-2)}}{5} \cos(t-2) - \frac{2e^{-(t-2)}}{5} \sin(t-2) \right] u(t-2)$$

solving ③

$$\mathcal{L}^{-1} \left\{ \frac{e^2 \cdot e^{-2s}}{s^2+2s+2} \right\} = e^2 \mathcal{L}^{-1} \left\{ \frac{e^{-2s}}{s^2+2s+2} \right\}$$

$$= e^2 [f(t-2) u(t-2)]$$

$$= e^2 [e^{-t} \sin(t-2) u(t-2)]$$

solving (4)

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 2s + 2} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2 + 1} \right\} = \underline{\underline{e^{-t} \sin t}}$$

$\therefore (1) + (2) + (3) + (4)$ gives,

$$\Rightarrow \frac{e^t}{5} - \frac{e^{-t}}{5} \cos t - \frac{2}{5} e^t \sin t - e^2 \left[\frac{e^{t-2}}{5} - \frac{e^{-(t-2)}}{5} \cos(t-2) - \frac{2}{5} e^{t-2} \sin(t-2) \right]$$

$$u(t-2) = e^2 [e^{-t} \sin(t-2) u(t-2)] + e^{-t} \sin t$$

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